Active and Passive Portfolio Management with Latent Factors

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June 22, 2026

1 Introduction

Ever since the pioneering work of Merton [Mer69], techniques from stochastic control theory have been applied extensively to solve dynamic portfolio optimization problems. Much of the work in this area of mathematical finance in the fifty years following Merton has been to develop market models and gain functionals which more closely resemble the types of portfolio optimization problems investment practitioners encounter in the real world.

In modern financial markets, there are a wide range of investment funds employing an even greater variety of strategies to meet the diverse needs of their clients. One of the most widely used characteristics for classifying investment strategies is the activity level involved in implementing the investment strategy. Broadly speaking, strategies, as well as the portfolios managers (PMs) which manage them, can be characterized as being either *active* or *passive*. While not strict binary classes by any means, the distinction between active and passive strategies is most clearly illustrated through the lens an investor's goals.

At a high level, every investor desires a portfolio which provides some combination of maximal returns and minimal risk. It has been known since at least Markowitz [Mar52] that in the absence of arbitrage, the superior returns investor hope to achieve–relative to the market portfolio–comes in the form of *risk premium*, i.e., compensation for incurring portfolio risk. Under this framework, active trading strategies consist of constructing portfolios of individuals securities meant to outperform the market, while minimizing the additional risk incurred. On the other hand, a portfolio is said to be passive if it is constructed to replicate the risk and return performance of the market. The goals of active and passive portfolio management are naturally formulated *relative* to the market (portfolio). Active portfolio management seeks to outperform the market, while passive portfolio management seeks to track the market.

In [AAJ18], Al-Aradi and Jaimungal consider a stochastic control problem (SCP) which formalizes the performance and tracking relative performance criteria that are ubiquitous in modern portfolio management. Their performance criterion incorporated both an outperformance benchmark and a tracking benchmark portfolio. The investor's desire for outperformance and tracking where encoded via the expectation and quadratic variation of relative log wealth of their portfolio against the respective benchmarks at terminal time. Al-Aradi and Jaimungal solved the resulting control problem using the standard dynamic programming approach of characterizing the associated value function via the Hamilton-Jacobi Bellman equation.

The closed-form expression of the optimal portfolio obtained in [AAJ18] relies on the growth rates of the market model. It is well known that asset growth rates are difficult to accurately estimate (see for example, [MLZ22]). To overcome this difficulty, Al-Aradi and Jaimungal built upon their previous work in [AAJ21], considering a general market model driven by latent asset growth rates and noise. The corresponding optimal portfolio in this new control problem is formulated in terms of filtered estimates of the market parameters, representing the investors' best guess of the market growth rates and covariance matrix given all available observable information. Using latent parameters also allowed Al-Aradi and Jaimungal to consider much more general growth rates; dropping the Markovian, smoothness, and deterministic assumptions imposed in [AAJ18]. In order to solve the optimal control problem in this level of generality, Al-Aradi and Jaimungal employ techniques from convex analysis in place of the classical HJB approach.

The goal of this expository work is two-fold:

(1) We reproduce the major theoretical results in [AAJ21], giving detailed arguments and outlining the necessary prerequisite knowledge so that the material can be understood by a first-year graduate student studying mathematics or statistics.

(2) We attempt to reproduce the numerical experiments carried out in [AAJ21], as well as extend the experimental findings of the performance of the optimal portfolio on previously unseen market data.

Regarding point (2), all of the relevant code to reproduce the figures and findings in this work can be accessed via the Github repository, github.com/brianceco/portfolio-management-latent-factors.

2 Literature Review

As mentioned in the introduction, there have been a variety of extensions to Merton's portfolio selection problem over the years, in various directions. Shortly after his original publication, Merton [Mer71] extended his work by considering more general utility functions. Transaction costs were incorporated by Davis and Norman in [DN90]. Browne [Bro97] introduced the need for the investor to make fixed withdrawals to pay off a liability, resulting in a positive probability of bankruptcy. In [BSEJM08], the authors introduce stochastic investment horizon. Liquidity constraints are considered in [APW14]. All of the above portfolio optimization problems consider absolute performance criteria, such as discounted utility or terminal wealth. The use of active performance criteria to encode active portfolio management objectives was introduced in [Bro99a] and [Bro99b] for a deterministic benchmark portfolio. In general, the absence of relative performance selection criteria represents a serious gap in the mathematical finance literature filled by [AAJ18] and [AAJ21].

The portfolio selection problem considered in both [AAJ18] and [AAJ21] was formulated within the stochastic portfolio theory (SPT) framework of Fernholz [Fer02]. SPT posits a mathematical model of equity markets which attempts to capture macroscopic properties [CSW24]. It is a natural setting in which to consider relative performance criteria in terms of benchmark portfolio. The authors were partly inspired by [Ode15], in which Odera solves a static portfolio selection problem in an SPT setting and using a relative performance criterion with respect to the market portfolio.

3 The Market Model

Throughout, we consider a portfolio optimization problem with a finite investment horizon $T > 0$, and we fix a natural number $n \in \mathbf{N}$ corresponding to the number of (risky) assets in our market model. We do not include the existence of a risk-free asset.

Convention 3.1 (Sample space)**.** We fix a filtered probability space $(\Omega, \mathbb{G}, \mathcal{G}, \mathbf{P})$, where $\mathbb{G} = (\mathcal{G}_t)_{t\in[0,T]}$ is the natural filtration generated by all processes in the model, augmented so as to be complete and right-continuous (i.e., to satisfy the usual conditions).

Convention 3.2 (Path Regularity)**.** We assume throughout that all processes are *right continuous with left limits* (RCLL) (aka *cádlág*), unless otherwise stated. With that said, we ignore, both in our notation and development of the theory, discrepancies between RCLL and continuous semimartingales. The relevant stochastic analysis needed for our purposes does not depend on this discrepancy and thus allows to proceed under this simplifying heuristic.

Definition 3.3 (Market model)**.** Given $i \in [n] = \{1, \dots, n\}$, the **stock price process** $X^i = (X^i_t)_{t\in[0,T]}$ for asset i is a positive semimartingale defined by

$$X^i_t = x^i_0 \exp\left(\int_0^t \gamma^i_s \, ds + M^i_t\right) \tag{3.4}$$

where $\gamma^i = (\gamma^i_t)_{t\in[0,T]}$ is a $\mathcal{G}$-progressive process called the asset's (**instantaneous**) **growth rate**, and $M^i = (M^i_t)_{t\in[0,T]}$ is a $\mathcal{G}$-progressive martingale with $M^i_0 = 0$, called the asset's **noise component**.

Remark 3.5. In SPT, one typically assumes for simplicity that each company in the market has a single share of stock outstanding, in which case the process X also represents the *market capitalizations* of stocks within the market.

Remark 3.6. The market model 3.3 considered in SPT is referred to as a *logarithmic model*, as it naturally lends itself to An application of Itô's lemma implies the n-dimensional process $\log X = (\log X^1, .., \log X^n)$ satisfies the SDE

$$\begin{cases} d\log X_t = \gamma_t dt + dM_t, & t \in [0,T] \\ X_0 = x_0 \end{cases}.$$

Assumption 1 (Integrability)**.** The growth rate γ and martingale M characterizing the market price process 3.4 will satisfy one of the following assumptions throughout:

(I1) We have $\gamma, M \in \mathbf{L}^2$.

(I2) We have $\gamma \in \mathbf{L}^{\infty,C}$ and $M \in \mathbf{L}^1$, where

$$\mathbf{L}^{\infty,C} := \left\{ \varphi \colon [0,T] \to \mathbf{R}^n : \varphi \text{ is } \mathbb{G}\text{-progressive and } \sup_{0\le s\le T} \|\varphi\|_\infty \le C \quad \mathbf{P}\text{-a.s.} \right\}$$

and given an integer $1 \le p < \infty$,

$$\mathbf{L}^p := \left\{ \varphi \colon [0,T] \times \Omega \to \mathbf{R}^n : \varphi \text{ is } \mathbb{G}\text{-progressive and } \|\varphi\|^p_{\mathbf{L}^p} := \mathbf{E}\left[\int_0^T \|\varphi_t\|_p^p \, dt \right] < \infty \right\}.$$

Definition 3.7. The **quadratic covariation** process $\Sigma : [0,T] \times \Omega \to \mathbf{R}^{2n}$ for the noise component M is defined by

$$\Sigma^{ij} := \frac{d}{dt} \langle \log X^i, \log X^j \rangle_t = \frac{d}{dt} \langle M^i, M^j \rangle, \quad 1 \le i,j \le n.$$

Assumption 2 (Covariation)**.** We assume the covariation process Σ is coercive and bounded, uniformly in t. That is, there exist constants $\varepsilon, C' > 0$, such that

$$\varepsilon |x|^2 \le x^\top \Sigma x \le C' |x|^2 \tag{3.8}$$

for all $x \in \mathbf{R}^n$ and $t \in [0,T]$.

Remark 3.9.

1. Unsurprisingly, we only need assumption 2 to hold for almost all paths $\omega \in \Omega$.

2. Some work is required to see that the right-hand inequality in 3.8 is equivalent to Σ being uniformly bounded. That the latter implies the former follows immediately from the Cauchy-Schwarz inequality. For the forward direction, we note that since Σ is symmetric, by the Spectral Theorem we have a decomposition

   $$\Sigma = UDU^\top$$

   where $U_t \in O(n)$ is a process valued in orthogonal $n \times n$ matrices, and $D_t = (D^{ii})$ is a process valued in diagonal matrices. Since Σ is positive semi-definite (in fact positive-definite by coercivity) and similar to D, using the right-hand inequality in 3.8 with $x = e_i$ the i^{th} standard basis vector for each $1 \le i \le n$, we conclude

   $$D^{ii} \le \mathrm{tr}(D) = \mathrm{tr}(\Sigma) \le nC', \quad 1 \le i \le n.$$

   But then given $x \in \mathbf{R}^n$,

   $$|\Sigma x|^2 = |UDU^\top x|^2 = |DU^\top x|^2 \le \max_{1\le i\le n} (D^{ii})^2 |U^\top x|^2 \le nC' |U^\top x|^2 = nC' |x|^2$$

   hence Σ is bounded uniformly in $t \in [0,T]$.

3. Importantly, the constant C in Assumption 1 can depend on the constants ε and C' in Assumption 2. After fixing such constants, there is a choice of C large enough such that the optimal portfolio we obtain is admissible under Assumption 2 (I2), as defined in the next subsection.

3.1 The observable filtration and (admissible) portfolios

As mentioned in the introduction, we consider a stochastic control problem in which the state process is driven by latent stochastic factors. In particular, the growth rate γ and noise M of the market price process X are unobservable to the portfolio manager. The PM must construct portfolios using only information within the filtration $\mathbb{F} = (\mathcal{F}_t)_{t\in[0,T]}$ given by the augmentation of the filtration generated by the market price process:

$$\mathcal{F}_t^X \coloneqq \sigma(\{X_s : s \le t\}), \quad t \in [0,T].$$

Definition 3.10. We call $\mathbb{F}$ the **observable filtration**. A process which is $\mathbb{F}$-adapted is said to be **observable**.

Somewhat informally, we say a process is *unobservable* if it is not observable in general without imposing additional assumptions. The most important example of observable processes are portfolios.

Definition 3.11. A **portfolio** $\pi = (\pi_t)_{t\in[0,T]}$ is an $\mathbb{F}$-progressive process which satisfies

$$\pi_t^\top \mathbf{1} = 1$$

for all $t \in [0,T]$. We let $\mathcal{P}$ denote the set of all portfolios.

We restrict the class of portfolios $\mathcal{P}$ which satisfy nice integrability properties, determined by which version of Assumption 2 we are considering.

Definition 3.12.

1. Under Assumption 2 (I1), the **class of admissible portfolios** $\mathcal{A}_2$, is defined by

$$\mathcal{A}_2 \coloneqq \mathcal{P} \cap \mathbf{L}^2.$$

2. Under Assumption 2 (I2), the **class of admissible portfolios** $\mathcal{A}_\infty$, is defined by

$$\mathcal{A}_\infty \coloneqq \mathcal{P} \cap \mathbf{L}^{\infty,C}.$$

When we are considering the class of admissible portfolios without specifying Assumption 2, we write $\mathcal{A}_\bullet$.

Remark 3.13. We note that the inclusion $\mathbf{L}^{\infty,C} \subset \mathbf{L}^\infty \subset \mathbf{L}^2$ of L^p spaces for finite measures (in this case, $([0,T]\times\Omega, \mathcal{B}([0,T])\otimes\mathcal{F}, dt\times d\mathbf{P})$) implies the inclusion of admissible spaces $\mathcal{A}_\infty \subset \mathcal{A}_2$. One way to interpret the distinction between assumptions (I1) and (I2) in Assumption 2 is that, if we allow our noise process M to be in $\mathbf{L}^1$, we must restrict the growth rates we allow (from $\mathbf{L}^2$ to $\mathbf{L}^{\infty,C}$), as well as the admissible portfolios we consider.

3.2 The value process of a portfolio

The definition of portfolios given by 3.11 is in terms of *weights*, i.e., the proportion of the portfolio's value attributable to the holdings in each stock, rather than the dollar holdings owned or owing. Given a portfolio $\pi \in \mathcal{P}$, let $\Delta = (\Delta_t)_{t\in[0,T]}$ denote the $\mathbf{R}^n$-valued process corresponding to the portfolio's position (i.e., number of shares owned or owing) in each stock over time. It follows that the portfolio's value process $V^\pi = (V_t^\pi)_{t\in[0,T]}$ is defined by

$$V_t^\pi \coloneqq \Delta_t^\top X_t, \quad t \in [0,T],$$

where $\Delta_t^i X_t^i$ denote the dollar holdings in stock i at time t, for each $i \in [n]$. Consequently, assuming $V^\pi > 0$ for each time $t \in [0,T]$ almost surely, we have the relation

$$\pi_t = \frac{\Delta_t * X_t^i}{V_t^\pi} \tag{3.14}$$

for each $t \in [0,T]$, where $x * y \coloneqq (x_i y_i)_{i=1}^n$ denotes the Hadamard product of $x, y \in \mathbf{R}^n$.

Over a significantly small period of time $\Delta t = [t, t+dt]$, over which the portfolio holdings Δ_t remain constant, the change dV_t in the value of the portfolio is given by the sum of the changes in the value of its individual holdings. Under this intuition, the dynamics of the value process V^π are characterized by the SDE

$$dV_t^\pi = \Delta_t^\top dX_t, \quad t \in [0,T].$$

We can then use Equation 3.14 to write an equivalent SDE characterizing the portfolio value process, which we take as the definition.

Definition 3.15. Given a portfolio $\pi \in \mathcal{A}_\bullet$, the associated **value** (or **wealth**) **process** $V^\pi = (V_t^\pi)_{t\geq 0}$ with initial wealth $z_0 >$ is the solution of the SDE

$$\begin{cases} dV_t^\pi = V_t^\pi \sum\limits_{i=1}^n \pi_t^i \frac{dX_t^i}{X_t^i}, \quad t \in [0,T] \\ V^\pi(0) = v_0 \end{cases}.$$

Under this model, we get the following characterization of the logarithm of the portfolio value process.

Proposition 3.16 ([Fer02], Proposition 1.1.5)**.** *The logarithm of the portfolio value process satisfies the SDEs*

$$d\log V_t^\pi = \gamma_t^\pi dt + \pi_t^\top dM_t \tag{3.17}$$

$$= \pi_t^\top d\log X_t + \Gamma_t^\pi \tag{3.18}$$

where

$$\gamma_t^\pi = \pi_t^\top \gamma_t + \Gamma_t^\pi, \quad \Gamma_t^\pi = \frac{1}{2}(\pi_t^\top \operatorname{diag}(\Sigma_t) - \pi_t^\top \Sigma_t \pi_t). \tag{3.19}$$

Definition 3.20. The processes γ_t^π and Γ_t^π in Proposition 3.16 are called the **portfolio growth rate** and **excess growth rate**, respectively.

4 The Stochastic Control Problem

4.1 Problem statement

We are now ready to define the optimal control problem which will be the subject of this report. Our guiding motivation is to incorporate the performance and tracking portfolio benchmarks which encode the investor's risk-return profile as optimization criteria. To that end, we fix two portfolios $\alpha \in \mathcal{A}_\bullet$ and $\beta \in \mathcal{A}_\bullet$, which serve as our *performance benchmark* and *tracking benchmark* respectively. These are taken as fixed–chosen according to the investor's preferences prior to portfolio construction. The gain functional for the control problem should incorporate both the desire of the investor to maximize return performance relative to α and minimize the deviation from β over the investment horizon T.

The main state variable $V^{\pi,\alpha} = (V_t^{\pi,\alpha})_{t\in[0,T]}$ we will use in the optimization problem is the (log) relative return of an arbitrary control portfolio $\pi \in \mathcal{A}_\bullet$ with respect to the performance benchmark α. The process is characterized by the SDE

$$d\log V_t^{\pi,\alpha} = d\log\left(\frac{V_t^\pi}{V_t^\alpha}\right) = d\log V_t^\pi - d\log V_t^\alpha = (\gamma_t^\pi - \gamma_t^\alpha)dt + (\pi_t - \alpha_t)^\top dM_t, \quad t \in [0,T]. \tag{4.1}$$

Since our portfolios are assumed to be in $\mathbf{L}^2$, we can integrate this SDE directly to obtain a strong solution

$$V_t^{\pi,\alpha} = V_0^{\pi,\alpha} + \int_0^t (\gamma_s^\pi - \gamma_s^\alpha)\, ds + \int_0^t (\pi_s - \alpha_s)\, dM_s, \quad t \in [0,T]. \tag{4.2}$$

Problem 4.3 (SCP)**.** The stochastic control problem is to find an *optimal portfolio* $\pi^* \in \mathcal{A}_\bullet$, characterized by

$$\pi^* = \operatorname*{argmax}_{\pi \in \mathcal{A}_\bullet} F(\pi), \tag{4.4}$$

in which $F\colon \mathcal{A} \to [-\infty, \infty]$ is the *performance criteria* (or *gain functional*), defined by

$$F(\pi) \coloneqq \mathbf{E}\left[\xi^0 V_T^{\pi,\alpha} - \frac{1}{2}\int_0^T \xi_s^1 (\pi_s - \beta_s)^\top \Omega_s (\pi_s - \beta_s)\, ds - \frac{1}{2}\int_0^T \xi_s^2 \pi_s^\top Q_s \pi_s\, ds\right], \tag{4.5}$$

where

- our unconstrained search space $\mathcal{A}$ is defined by

$$\mathcal{A} \coloneqq \left\{\pi \in \mathbf{L}^2 : \pi \text{ is } \mathbb{F}\text{-progressive}\right\};$$

- $\xi^0 \in \mathbf{R}_{\geq 0}$ is a constant, $\xi^1 = (\xi_t^1)_{t\in[0,T]}$, and $\xi^2 = (\xi_t^2)_{t\in[0,T]}$ are $\mathbb{F}$-progressive bounded $\mathbf{R}_{\geq 0}$-valued processes such that $\xi^0 + \xi^1 + \xi^2$ is uniformly bounded away from 0;
- $\Omega = (\Omega_s)_{s\in[0,T]}$ and $Q = (Q_s)_{s\in[0,T]}$ are $\mathbf{R}^{2n}$-valued $\mathbb{F}$-progressive covariance processes, i.e., for every $s \in [0,T]$, Ω_s and Q_s are random variables valued in positive semi-definite symmetric matrices.

Remark 4.6 (Interpreting the performance criterion)**.** The summands making up the performance criterion 4.5 can be understood as follows:

1. (*Terminal outperformance*). The first term is a terminal (i.e., path independent) reward which captures the investor's desire to maximize the expected relative outperformance against the outperformance of their portfolio against the benchmark α at terminal time. It can be interpreted as maximizing the expected utility of relative wealth under a log utility $U(x) = \log x$.
2. (*Running tracking error*). The second term is a running penalty given by the covariation of the difference $\pi - \beta$ with respect to the (instantaneous) covariation process Ω. It captures the investor's aversion to *relative risk*, i.e., it encodes their desire for their portfolio returns to track those of the benchmark β. In the case $\Omega = \Sigma$, it reduces to the covariation of $\log V^{\pi,\beta}$.
3. (*Running absolute risk*). The third term is a running penalty given by the covariation of the portfolio with respect to the covariation process Q. It captures the investor's aversion to *absolute risk*. In the case $Q = \Sigma$, it reduces to the covariation of $\log V^\pi$. One can also take $Q = \operatorname{diag}(p)$, where $p \in [0,\infty)^n$ is a weight vector which associates penalty p^i to holding asset i for each $i \in [n]$.
4. (*Investor preference parameters*). The constant ξ^0, as well as the processes ξ^1 and ξ^2, are meant to capture the particular risk-return preferences of the investor. In particular, ξ^0, ξ^1, and ξ^2 capture the importance the investor places on relative outperformance, tracking error, and absolute risk, respectively. These scaling factors re-weight the respective summands in the expectation 4.5, producing a personalized performance criteria $F(\pi)$ which better optimizes for their goals. By the boundedness assumptions, we can assume ξ^0, ξ^1, and ξ^2 take values in a common interval $[0, \xi^U]$ for some fixed $0 < \xi^U < \infty$. We let $\xi^L > 0$ denote a lower bound (uniform in $[0,T]$) for $\xi^0 + \xi^1 + \xi^2$.

Remark 4.7 (Existence and uniqueness of an optimal portfolio)**.** Of course, as is typical in stochastic control theory, a priori the existence and uniqueness of an optimal portfolio defined by 4.4 is not at all guaranteed. The main result of [AAJ21] is to show that for the particular gain functional defined by 4.5, a unique optimal portfolio π^* does in fact exist. Furthermore, such an optimal portfolio is admissible in that $\pi^* \in \mathcal{A}_\bullet \subset \mathcal{A}$, depending which of Assumption 2 we are considering. Since the optimal portfolio is obtained via methods other than the DPP, we have no need to let the gain functional depend on the initialization parameters given by initial wealth $x \in \mathbf{R}^n$ or starting time $t \in [0,T]$. We also do not need to consider the value function which is characteristic of a stochastic control problem.

We impose restrictions on the relative and absolute penalty matrix processes similar to that imposed on the covariation process Σ in Assumption 2.

Assumption 3 (Relative and absolute penalties)**.** The processes Ω and Q are uniformly coercive and bounded. That is, there exists constants $\varepsilon, C' > 0$ (which we can take to be the same as in 2) such that

$$\varepsilon|x|^2 \leq x^\top \Omega x \leq C'|x|^2 \quad \text{and} \quad \varepsilon|x|^2 \leq x^\top Q x \leq C'|x|^2$$

for all $x \in \mathbf{R}^n$

4.2 Simplifications of the control problem

We begin our analysis of the stochastic control problem 4.3 by making some simplifications to the performance criteria 4.5.

Lemma 4.8. *Under either assumption 1 (I1) or (I), the stochastic integral*

$$I_t \coloneqq \int_0^t (\pi_s - \alpha_s)^\top \, dM_s, \quad t \in [0,T]$$

appearing in the relative wealth process 4.2 is a (true) $(\mathbb{G}, \mathbf{P})$*-martingale.*

Proof. A priori we know I is a local martingale such that $I \in L^2_{\text{loc}}(d\langle M \rangle)$. By the Itô isometry, we have

$$\begin{aligned}\mathbf{E}[|I_T|^2] &= \mathbf{E}[\langle I_T \rangle] \\ &= \mathbf{E}\left[\int_0^T (\pi_s - \alpha_s)^\top \Sigma_s (\pi_s - \alpha_s)\, ds\right] \\ &\leq C'(\|\pi\|_{\mathbf{L}^2} + \|\alpha\|_{\mathbf{L}^2})\end{aligned}$$

.

Thus we conclude I is a (uniformly integrable) martingale. ☐

In light of Lemma 4.8, assuming $V_0^\pi = V_0^\alpha$, we can rewrite the performance criteria as

$$F(\pi) = \mathbf{E}\left[\int_0^T \xi^0(\gamma_s^\pi - \gamma_s^\alpha) - \frac{1}{2}\xi_s^1(\pi_s - \beta_s)^\top \Omega(\pi_s - \beta_s) - \frac{1}{2}\xi_s^2 \pi_s^\top Q_s \pi_s \, ds\right]. \tag{4.9}$$

As is currently defined (by equation 4.9), the performance criteria F depends on unobservable (i.e., $\mathbb{G}$-progressive) processes, namely the market growth rate γ and the market covariation Σ. In order to be able to actually implement any optimal control we obtain, we need to reformulate the control problem in terms of observable processes.

Definition 4.10. Given a $\mathbb{G}$-adapted process $\varphi : [0,T] \times \Omega \to \mathbf{R}^n$, the **filter** $\widehat{\varphi} = (\widehat{\varphi}_t)_{t\in[0,T]}$ is the $\mathbb{F}$-adapted $\mathbf{R}^n$-valued process defined by

$$\widehat{\varphi}_t \coloneqq \mathbf{E}[\varphi | \mathcal{F}_t], \quad t \in [0,T],$$

where the conditional expectation is taken entrywise.

We would like the performance criteria F to be defined as an expectation taken with respect to the observable probability space $(\Omega, \mathbb{F}, \mathcal{F}, \mathbf{P})$. We consider the *filtered growth rate* $\widehat{\gamma} = (\gamma_t)_{t\in[0,T]}$ and *filtered covariation* $\widehat{\Sigma} = (\widehat{\Sigma}_t)_{t\in T}$. These are $\mathbb{F}$-progressive processes representing the portfolio manager's best prediction for their respective unobservable quantities, given all available stock price information. We note some properties of these filtered process.

Proposition 4.11.

1. *The filtered covariation process* $\widehat{\Sigma}$ *is coercive and bounded, uniformly in* t*, as in Assumption 2.*
2. *The innovations process* $\widehat{M} = (\widehat{M}_t)_{t\in[0,T]}$ *defined by*

$$\widehat{M}_t \coloneqq \log X_t - \int_0^t \widehat{\gamma}_s \, ds, \quad t \in [0,T]$$

is an $\mathbb{F}$*-progressive martingale with* $\widehat{M} \in \mathbf{L}^1$*. Furthermore, the logarithm of asset prices satisfies the observable SDE*

$$d \log X_t = \widehat{\gamma}_t dt + d\widehat{M}_t, \quad t \in [0,T].$$

Proof. For the first statement, we note that Assumption 2 implies the entries $\Sigma^{ij} = \langle M^i, M^j \rangle$ of Σ are bounded uniformly in $t \in [0,T]$. Indeed, by Remark 3.9 (2), Σ is uniformly bounded as a (stochastic) matrix, hence there exists $C'' > 0$ such that, for each $1 \leq i, j \leq n$, we have

$$(\Sigma^{ij})^2 \leq \sum_{i=1}^{n} (\Sigma^{ij})^2 = |\Sigma e_j|^2 \leq C.$$

This implies $\widehat{\Sigma}$ also has uniformly bounded entries. Uniform coercivity follows from the fact that conditional expectation is order-preserving, and that given $x \in \mathbf{R}^n$, $\varepsilon |x|^2$ is a constant, and hence agrees with its conditional expectation.

For the innovations process, we begin by proving that $\widehat{M} \in \mathbf{L}^1$. Firstly, we claim $\log X \in \mathbf{L}^1$. We have

$$\begin{aligned} \mathbf{E}\left[\int_0^T \left|\int_0^t \gamma_s \, ds\right| dt\right] &= \mathbf{E}\left[\int_0^T \sum_{i=1}^{n} \left|\int_0^t \gamma_s^i \, ds\right| dx\right] \\ &\leq \mathbf{E}\left[\int_0^T \int_0^t |\gamma_s| \, ds \, dt\right] \\ &\leq T \|\gamma\|_{\mathbf{L}^1} \end{aligned}$$

Since $\gamma \in \mathbf{L}^1$ regardless of which part of Assumption 1 we assume, the result follows. That $\widehat{M} \in \mathbf{L}^1$ then follows by repeating the above computation with $\widehat{\gamma}_t$ replacing γ_t, together with the conditional Fubini theorem.

As a sum of $\mathbb{F}$-progressive process, we also note $\widehat{M}$ itself is $\mathbb{F}$-progressive. Finally, given $0 \leq s \leq t \leq T$,

$$\begin{aligned} \mathbf{E}[\widehat{M}_t | \mathcal{F}_s] &= \mathbf{E}\left[\int_0^t (\gamma_u - \widehat{\gamma}_u) \, du + M_t \middle| \mathcal{F}_s\right] \\ &= \int_0^t \mathbf{E}[\gamma_u - \widehat{\gamma}_u | \mathcal{F}_s] \, du + \mathbf{E}[M_t | \mathcal{F}_s] && \text{(conditional Fubini theorem)} \\ &= \mathbf{E}[M_t | \mathcal{F}_s] \\ &= \mathbf{E}[M_s | \mathcal{F}_s] && \text{(Tower law)} \\ &= \mathbf{E}\left[\log X_s - \int_0^t \gamma_u \, du \middle| \mathcal{F}_s\right] \\ &= \log X_s - \int_0^t \widehat{\gamma}_u \, du && \text{(conditional Fubini theorem)} \\ &= \widehat{M}_s \end{aligned}$$

hence $\widehat{M}$ is an $\mathbb{F}$-martingale. □

In order to express the performance criteria 4.9 in terms of the filtered growth rate and covariation, we notice that we only need to modify the first summand. Noting that the portfolios π and α are $\mathbb{F}$-progressive, ξ^0 is constant, and that $\mathbf{E}[\gamma_t] = \mathbf{E}[\widehat{\gamma}_t]$ and $\mathbf{E}[\Sigma_t] = \mathbf{E}[\widehat{\Sigma}_t]$, by Fubini's theorem we have

$$F(\pi) = \mathbf{E}\left[\int_0^T \xi^0(\widehat{\gamma}_s^\pi - \widehat{\gamma}_s^\alpha) - \frac{1}{2}\xi_s^1(\pi_s - \beta_s)^\top \Omega_s (\pi_s - \beta_s) - \frac{1}{2}\xi_s^2 \pi_s^\top Q_s \pi_s \, ds\right]$$

where the filters $\widehat{\gamma}^\pi$ and $\widehat{\gamma}^\alpha$ are defined as in 3.19. We then have

$$F(\pi) = \mathbf{E}\left[\int_0^T \xi^0 \widehat{\gamma}_s^\pi - \frac{1}{2}\xi_s^1 \pi_s^\top \Omega \pi_s + \xi_s^1 \pi_s^\top \Omega_s \beta_s - \frac{1}{2}\xi_s^2 \pi_s^\top Q_s \pi_s \, ds\right] - \mathbf{E}\left[\int_0^T \xi^0 \widehat{\gamma}_s^\alpha + \frac{1}{2}\xi_s^1 \beta_s^\top \Omega_s \beta_s \, ds\right]. \tag{4.12}$$

As the second expectation does not depend on the control portfolio π, we can disregard it without altering the optimal portfolio. Expanding the filtered portfolio growth rate and grouping like terms, we obtain

$$F(\pi) = \mathbf{E}\left[\int_0^T -\frac{1}{2}\pi_s^\top A_s \pi_s + \pi_s^\top B_t \, ds\right] \tag{4.13}$$

where the $\mathbf{R}^{n^2}$- and $\mathbf{R}^n$-valued processes $A = (A_t)_{t\in[0,T]}$ and $B = (B_t)_{t\in[0,T]}$ are defined by

$$A_t \coloneqq \xi^0\widehat{\Sigma}_t + \xi_t^1\Omega_t + \xi_t^2 Q_t \quad \text{and} \quad B_t = \xi^0\widehat{\gamma}_t + \frac{1}{2}\mathrm{diag}(\widehat{\Sigma}_t) - \xi_t^1\Omega_t\beta_t, \quad t \in [0,T]. \tag{4.14}$$

4.3 Finding the optimal portfolio via convex analysis

We begin our search for an optimal portfolio π^* solving the stochastic control problem 4.3. If Assumption 2 (I1) or (I2) is enforced, our search space shrinks to the space of admissible portfolios $\mathcal{A}_2$ and $\mathcal{A}_\infty$ respectively. We begin by noting that $\mathcal{A}$–being a genuine L^2 space on $([0,T]\times\Omega, dt\times d\mathbf{P})$ with the $\mathbf{F}$-progressive σ-algebra– is a reflexive Banach space ([SS11], Theorem 4.1). We let $\mathcal{A}^*$ denote the corresponding dual space, and denote by $\langle -,-\rangle$ the canonical bilinear pairing on $\mathcal{A}\times\mathcal{A}^*$. Our goal will be to analyze the functional F through the convex-analytic framework outlined in Appendix A.

Proposition 4.15. *The performance criteria F defined by 4.13 is strictly concave, upper semicontinuous, and proper.*

Proof. We begin by proving F is strictly concave. Note that the matrix process A_t appearing in 4.14 is positive definite at all times $t \in [0,T]$ by coercivity (and the fact that the preference parameters ξ^i are nonnegative), hence the associated quadratic form map $q_{A_t} : \mathbf{R}^n \ni x \mapsto \frac{1}{2}x^\top A_t x$ is strictly concave. Furthermore, we have

$$F(\pi) = \mathbf{E}\left[\int_0^T q_{A_s}(\pi)\,ds\right] + L(\pi)$$

where $L\colon \mathcal{A} \to \mathbf{R}$ is a linear map. Thus, given $\tau \in (0,1)$ and $\pi, \widetilde{\pi} \in \mathcal{A}$, we have

$$\begin{aligned} F(\tau\pi + (1-\tau)\widetilde{\pi}) - \tau F(\pi) - (1-\tau)F(\widetilde{\pi}) &= \mathbf{E}\left[\int_0^T q_{A_s}(\tau\pi_s + (1-\tau)\widetilde{\pi}_s) - \tau q_{A_s}(\pi_s) - (1-\tau)q_{A_s}(\widetilde{\pi}_s)\,dx\right] \\ &> 0. \end{aligned}$$

That F is upper semicontinuous follows from Fatou's lemma. Indeed, given $\pi^{(n)} \to \pi$ in $\mathcal{A}$, by Hölder's inequality, $\{\pi^{(n)}\}$ is bounded in $\mathbf{L}^1$, and consequently,

$$\limsup_n F(\pi^{(n)}) \leq F(\limsup_n \pi^{(n)}) = F(\pi).$$

This same argument replacing Fatou's lemma with dominated convergence implies F is in fact continuous on $\mathcal{A}$. Finally, we show F is proper, i.e., F never attains ∞ and is somewhere greater than $-\infty$. By the uniform boundedness of the covariance processes $\widehat{\Sigma}$, Ω, and Q, and the fact that the growth rate γ is in $\mathbf{L}^2$, there exists a constant $C = C(C', \bar{\xi})$ such that, for any $\pi \in \mathcal{A}$,

$$\begin{aligned} |F(\pi)| &\leq C\mathbf{E}\left[\int_0^T |\pi_s|^2 + |B_s|^2\,ds\right] \\ &= C(\|\pi\|_{\mathbf{L}^2}^2 + \|\gamma\|_{\mathbf{L}^2}^2 + \|\beta\|_{\mathbf{L}^2}) \\ &< \infty \end{aligned}$$

completing the proof. ☐

Theorem 4.16 (Existence of Gâteux differential)**.** *Given $\pi \in \mathcal{A}$, the Gâteaux differential $F'(\pi)$ of F at π exists, and is given by*

$$\langle \widetilde{\pi}, F'(\pi)\rangle = F'(\pi;\widetilde{\pi}) = \mathbf{E}\left[\int_0^T \widetilde{\pi}^\top(-A_t\pi + B_t)\,dt\right], \quad \widetilde{\pi} \in \mathcal{A}. \tag{4.17}$$

Proof. Firstly, given $\pi \in \mathcal{A}$, it is clear the map $F'(\pi)$ given by Equation 4.17 is a bounded linear functional on $\mathcal{A}^*$. Linearity follows from the fact that F' is a composite of linear functions. Continuity follows from dominated convergence as in the proof of Proposition 4.15.

By Definition A.8, given $\pi, \widetilde{\pi} \in \mathcal{A}$, the directional derivative $F(\pi; \widetilde{\pi})$ of F at π in the $\widetilde{\pi}$ direction is given by

$$\begin{aligned} F(\pi;\widetilde{\pi}) &= \lim_{h\to 0^+} \frac{F(\pi + h\widetilde{\pi}) - F(\pi)}{h} \\ &= \lim_{h\to 0^+} \mathbf{E}\left[\int_0^T -\frac{h}{2}\widetilde{\pi}_s^\top A_s \widetilde{\pi}_s - \widetilde{\pi}_s^\top A \pi_s + \widetilde{\pi}_s^\top B_s \, ds\right]. \end{aligned}$$

Again, since $\widetilde{\pi}, \gamma \in \mathbf{L}^2$ and A is uniformly bounded, by dominated convergence we can pass the limit inside the integral to conclude

$$\begin{aligned} F(\pi;\widetilde{\pi}) &= \mathbf{E}\left[\int_0^T -\widetilde{\pi}_s^\top A \widetilde{\pi}_s + \widetilde{\pi}_s^\top B_s \, ds\right] \\ &= \mathbf{E}\left[\int_0^T \widetilde{\pi}_s^\top(-A\widetilde{\pi}_s + B_s)\, ds\right] \\ &= \langle \widetilde{\pi}, F'(\pi)\rangle \end{aligned}$$

hence $F'(\pi)$ is the Gâteux differential of F at π as required. □

In light of Proposition 4.15 and Theorem 4.16, we can check for an optimal portfolio via the first-order conditions given by Proposition A.11 (where we replace F with $-F$ to search for a global maximum). The next theorem gives an explicit construction of the unique optimal control.

Theorem 4.18 (Optimal solution to 4.3)**.** *The portfolio $\pi^* \in \mathcal{A}_\bullet$ defined by*

$$\pi^* := A^{-1}\left[\frac{1 - \mathbf{1}^\top A^{-1} B}{\mathbf{1}^\top A^{-1}\mathbf{1}}\mathbf{1} + B\right]$$

is the unique solution to the stochastic control problem 4.3.

Proof. We begin by verifying that, under either Assumption 2 (I1) or (I2), the portfolio π^* is admissible. To this end, we first prove that $\mathcal{A}_\bullet$ is a closed convex subset of $\mathcal{A}$. Convexity is clear: given two admissible portfolios $\pi, \widetilde{\pi} \in \mathcal{A}_\bullet$ and $\tau \in [0,1]$, we have

$$\mathbf{1}^\top(\tau\pi + (1-\tau)\widetilde{\pi}) = \tau + (1-\tau) = 1.$$

Furthermore, if $\bullet = 2$ i.e., we are in Assumption 2 (I1), then $\tau\pi + (1-\tau)\widetilde{\pi} \in \mathbf{L}^2$ since this is a vector space. If $\bullet = \infty$, we have

$$\sup_{t\in[0,T]} (\tau\pi_t + (1-\tau)\widetilde{\pi}_t) \le \tau \sup_{t\in[0,T]} \pi_t + (1-\tau) \sup_{t\in[0,T]} \widetilde{\pi}_t \le \tau C + (1-\tau)C = C \quad \mathbf{P}\text{-a.s.},$$

hence $\tau\pi + (1-\tau)\widetilde{\pi} \in \mathbf{L}^{\infty,C}$.

To prove $\mathcal{A}_\bullet$ is closed, suppose $\pi^{(n)} \in \mathcal{A}_\bullet$ converges to π in $\mathcal{A}$. In either case, we have by the triangle inequality,

$$\begin{aligned} \|\mathbf{1}^\top\pi - 1\|_{\mathbf{L}^2} &\le \|\mathbf{1}^\top(\pi - \pi^{(n)})\|_{\mathbf{L}^2} + \underbrace{\|\mathbf{1}^\top\pi^{(n)} - 1\|_{\mathbf{L}^2}}_{=0} \\ &\le \|\pi - \pi^{(n)}\|_{\mathbf{L}^2}^2 \\ &\to 0 \end{aligned}$$

as $n \to \infty$. If $\bullet = 2$, we immediately conclude $\pi \in \mathcal{P} \cap \mathbf{L}^2 = \mathcal{A}_2$, hence $\mathcal{A}_2$ is closed. Now, supposing $\bullet = \infty$, it remains to prove $\pi \in \mathbf{L}^{\infty,M}$. Let $A \subset \Omega$ be the subset of paths $\omega \in \Omega$ such that $\|\pi_t(\omega)\|_\infty > M$ for some $t \in [0,T]$. By right continuity, it follows that $|\{t \in [0,T] : \|\pi_t(\omega)\|_\infty > M\}| > 0$. We have

$$\begin{aligned}
\mathbf{E}\left[\mathbf{1}_A \int_0^T \|\pi_t\|_\infty - M)\, dt\right] &\leq \mathbf{E}\left[\mathbf{1}_A \int_0^T (\|\pi_t - \pi_t^{(k)}\|_\infty\, dt\right] + \underbrace{\mathbf{E}\left[\mathbf{1}_A \int_0^T (\|\pi_t^{(k)}\|_\infty - M)\, dt\right]}_{\leq 0} \\
&\leq \mathbf{E}\left[\mathbf{1}_A \int_0^T \|\pi_t - \pi_t^{(k)}\|_2\, dt\right] \\
&\leq T\|\pi - \pi^{(k)}\|_{\mathbf{L}^2}
\end{aligned}$$

where the last inequality follows by Hölder's inequality in $\mathbf{L}^2$. Taking $k \to \infty$, we obtain

$$\mathbf{E}\left[\mathbf{1}_A \int_0^T (\|\pi_t\|_\infty - M)\, dt\right] = 0$$

and since the integrand is positive, it follows that $\mathbf{P}(A) = 0$, and we conclude $\pi \in \mathcal{A}_\infty$ as required.

We now move on to proving π^* is in fact optimal. We begin by showing $\pi^* \in \mathcal{A}_\bullet$ is an admissible portfolio. Clearly π^* is $\mathbb{F}$-progressive as a linear combination of $\mathbb{F}$-progressive processes. We have

$$\begin{aligned}
\mathbf{1}^\top \pi^* &= \mathbf{1}^\top A^{-1}\left(\frac{1 - \mathbf{1}^\top A^{-1} B}{\mathbf{1}^\top A^{-1}\mathbf{1}} + B\right) \\
&= (1 - \mathbf{1}^\top AB) + \mathbf{1}^\top A^{-1} B \\
&= 1
\end{aligned}$$

hence $\pi^* \in \mathcal{P}$ is a portfolio in the sense of Definition 3.11.

To verify admissibility, first recall $A_t = \xi_0 \widehat{\Sigma}_t + \xi_t^1 \Omega_t + \xi_t^2 Q_t$. The preference processes ξ^i are bounded above by ξ^U, and the sum is bounded below by ξ^L (Remark 4.6 (4)). Since the covariation processes Σ, Ω, and Q are all uniformly bounded and coercive, it follows that A is itself uniformly bounded and coercive, and consequently, so is its inverse A^{-1}. Consequently, there exists a constant $C^* = C^*(C', \varepsilon, n, \xi^L, \xi^U)$ such that

$$\begin{aligned}
|\pi_t^*| &\leq A_t^{-1}\left(\frac{1 - \mathbf{1}^\top A_t^{-1} B_t}{\mathbf{1}^\top A_t^{-1}\mathbf{1}}\mathbf{1} + B_t\right) \\
&\leq C^*|B_t| \\
&= C^*|\xi^0 \widehat{\gamma}_t + \frac{1}{2}\operatorname{diag}(\widehat{\Sigma}_t) - \xi_t^1 \Omega_t \beta_t|.
\end{aligned}$$

Since ξ^0 and ξ^1 are bounded above (the latter uniformly in $t \in [0,T]$), $\widehat{\Sigma}_t$ and Ω are uniformly bounded, it is clear that establishing $\pi^* \in \mathcal{A}_\bullet$ reduces to $\widehat{\gamma} \in \mathbf{L}^2$ if $\bullet = 2$ or $\widehat{\gamma} \in \mathbf{L}^{\infty,C}$ if $\bullet = \infty$.

Finally, we establish optimality of the control π^* via the first-order conditions on the performance criteria given by Proposition A.11. Namely, given another admissible portfolio $\pi \in \mathcal{A}_\bullet$, we have

$$\begin{aligned}
\langle \pi - \pi^*, F'(\pi^*)\rangle &= \mathbf{E}\left[\int_0^T (\pi_t - \pi_t^*)^\top(-A_t \pi_t^* + B_t)\, dt\right] \\
&= \mathbf{E}\left[\int_0^T -\frac{1 - \mathbf{1}^\top A_t B_t}{\mathbf{1}^\top A_t \mathbf{1}}(\pi_t - \pi_t^*)^\top \mathbf{1}\, dt\right] \\
&= 0,
\end{aligned}$$

with the final equality following from the fact that $(\pi - \pi^*)^\top \mathbf{1} = 0$ as $\pi, \pi^* \in \mathcal{P}$ are portfolios. This completes the proof that π^* is the unique maximizer of the performance criteria F. $\square$

4.4 Understanding the optimal portfolio in terms of growth- and risk-optimality

The stochastic control portfolio problem considered in 4.3 is sufficiently general that we can use the optimal control 4.18 to construct other portfolios of interest as optimal portfolios under specific investor preferences.

Notation 4.19. Because of its prevalence in this section, for notational convenience, we define the $\mathbb{F}$-progressive process $\widehat{\alpha} = (\widehat{\alpha}_t)_{t\in[0,T]}$ by

$$\widehat{\alpha}_t := \xi^0\gamma_t + \frac{1}{2}\mathrm{diag}(\widehat{\Sigma}_t).$$

Thus, we can rewrite the process B used in defining the performance criteria 4.13 and the optimal portfolio 4.18 as

$$B_t = \widehat{\alpha}_t - \xi_t^1\Omega\beta_t, \quad t \in [0,T].$$

Definition 4.20.

1. A portfolio $\pi^{\mathrm{GOP}} \in \mathcal{A}_\bullet$ is **growth optimal** (or is a **GOP**) if

$$\pi^{\mathrm{GOP}} \in \operatorname*{argmax}_{\pi\in\mathcal{A}_\bullet} \mathbf{E}\left[\log V_T^\pi\right]. \tag{4.21}$$

2. A portfolio $\pi^{\mathrm{MQP}} \in \mathcal{A}_\bullet$ has **minimal quadratic variation** (or is an **MQP**) if

$$\pi^{\mathrm{MQP}} \in \operatorname*{argmin}_{\pi\in\mathcal{A}_\bullet} \mathbf{E}\left(\int_0^T \pi_t^\top\Sigma_t\pi_t\, dt\right) = \operatorname*{argmin}_{\pi\in\mathcal{A}_\bullet} \mathbf{E}\langle V^\pi\rangle_T. \tag{4.22}$$

A GOP maximizes absolute return–as measured by the expected terminal logarithm of returns–while a MQP minimizes absolute risk–as measured by the expected quadratic variation over the investment horizon.

Theorem 4.23 (Existence and uniqueness of GOP and MQP)**.** *In the market model 3.3, the minimum quadratic variation portfolio π^{MQP} and the growth optimal portfolio π^{GOP} both exist, and are uniquely defined by*

$$\pi^{\mathrm{MQP}} := \frac{1}{\mathbf{1}^\top\widehat{\Sigma}^{-1}\mathbf{1}}\widehat{\Sigma}^{-1}\mathbf{1} \tag{4.24}$$

and

$$\pi^{\mathrm{GOP}} := (1 - \mathbf{1}^\top\widehat{\Sigma}\widehat{\alpha})\pi^{\mathrm{MQP}}. \tag{4.25}$$

Proof. One can readily see that the stochastic control problem defining π^* (Equation 4.4) reduces to the minimum quadratic variation portfolio 4.22 by taking $\xi^0 = 0$, $\xi^1 \equiv 0$, $\xi^1 \equiv 1$, and $Q = \Sigma$. Consequently, $B = 0$ and A reduces to $\widehat{\Sigma}$ in 4.18 to give the MQP 4.24. Similarly, taking $\xi^1 \equiv 0 \equiv \xi^2$ reduces 4.4 to the growth optimal portfolio 4.21. In this case, $B = \widehat{\alpha}$ and A again reduces to $\widehat{\Sigma}$, yielding the desired form 4.25. □

5 An Extended Example: The hidden Markov market model

5.1 Filtering theory and HMMs

We will exhibit the flexibility and robustness of the stochastic control problem 4.3 by considering the case in which the market price process X (Definition 3.3) is given by a *hidden Markov model* (*HMM*).

Before specifying the specific market model we consider, because the study of HMMs will take us into filtering theory proper, we begin with formalizing the framework that we will be working within.

A *filtering problem* on the filtered probability space $(\Omega, \mathcal{G}, \mathbb{G}, \mathbf{P})$ consists of the following data:

1. A $\mathbb{G}$-adapted process $\Theta = (\Theta_t)_{t\in[0,T]}$ taking values in a (Polish) *state space* $\mathcal{S}$, which we call the *signal process*.

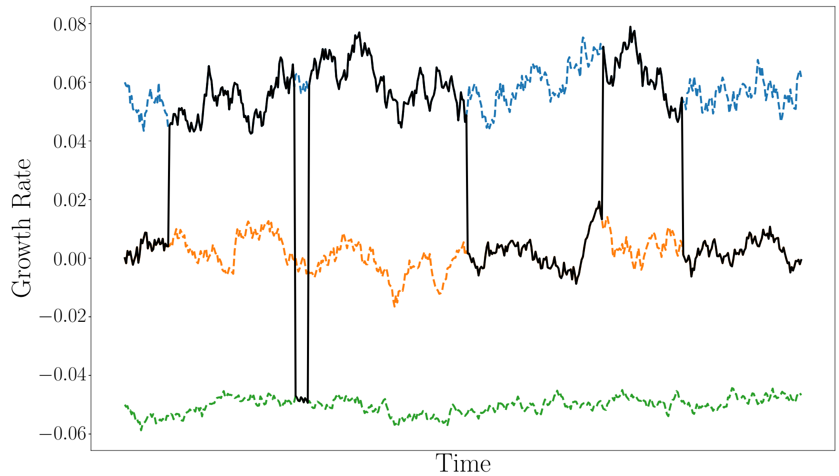

Figure 1: An example of a growth rate process $\gamma^{(\Theta)}$ under a Hidden Markov Model. The growth rate switches between three distinct OU processes $\gamma_t^{(i)}$–depicted in the figure above by the dashed paths–according to the state value of a latent Markov chain Θ_t. We can think of Θ as capturing different market regimes, for example, a bull market (blue), a neutral market (yellow), and a bear market (green).

2. An n-dimensional Itô process $Y = (Y_t)_{t\in[0,T]}$ of the form

$$Y_t := \int_0^t h(\Theta_s)\, ds + W_t, \quad t \in [0, T]$$

where W is an n-dimensional $(\mathbb{G}, \mathbf{P})$-Brownian motion independent of Θ, and $h \colon \mathcal{S} \to \mathbf{R}^n$ is a measurable map called the *observation function*. We call Y the *observation process*.

Letting $\mathbb{G}$ and $\mathbb{F}$ denote the augmentations of the canonical σ-algebras associated to the processes Θ and W.

At a high level, a *hidden Markov models* (HMMs) are a general class of models used extensively in filtering theory, in which the observation process Y and the signal process Θ have a joint Markovian structure, formalized as follows.

Specifically, we assume the logarithm market price process is given by

$$\begin{cases} d\log X_t = \gamma\Theta_t dt + \xi dW_t, & t \in [0, T], \\ \log X_0 = \log x_0 \end{cases} \tag{5.1}$$

where

- the signal process $\Theta = (\Theta_t)_{t\in[0,T]}$ is a homogeneous Markov chain with finite state space $E_K := \{e_1, \dots, e_k\}$ the set of standard basis vectors for $\mathbf{R}^k$, generator matrix G, and initial distribution π;
- the process $W = (W_t)_{0\le t\le T}$ is a standard n-dimensional Brownian motion assumed to be independent of Θ;
- the matrix $\gamma \in \mathbf{R}^{n\times k}$ is deterministic, homogeneous, and its columns

$$\gamma^{(k)} := \gamma e_k, \quad k \in [K],$$

represent market growth rates under the various states of Θ;
- the diffusion matrix $\xi \in \mathbf{R}^{n\times d}$ is assumed to be deterministic, homogeneous, and full rank,
- $x_0 \in \mathbf{R}^n_{>0}$ is a constant denoting the initial stock prices.

Abuse of Notation 5.2. We write $\gamma^{(\Theta_t)} \coloneqq \gamma\Theta_t$ to denote the growth rate process under the hidden Markov market model 5.1.

Remark 5.3 (Verification of model assumptions)**.**

1. It is clear that the market growth rate $\gamma^{(\Theta_t)}$ lies in $\mathbf{L}^{\infty,C}$ and the market noise component ξW lies in $\mathbf{L}^2$, hence Assumption 1 (I1) and (I2) both hold.
2. The quadratic covariation process of the market model is $\Sigma = \xi\xi^\top$, which is clearly bounded. Since ξ has full rank, Σ is also coercive, hence 2 is satisfied.

In order to discuss the optimal portfolio for the hidden Markov model, we consider the filtered growth rate

$$\widehat{\gamma}^{\Theta_t} \coloneqq \mathbf{E}[\gamma^{\Theta_t}|\mathcal{F}_t] = \gamma\widehat{\Theta}_t = \sum_{k=1}^{K} \gamma^{(k)}\mathbf{P}(\Theta_t = e_k|\mathcal{F}_t), \quad t \in [0,T].$$

For each state $e_k \in E_K$, we define the filter $p^k = (p_t^k)_{t\in[0,T]}$ by

$$p_t^k \coloneqq \mathbf{P}(\Theta_t = e_k|\mathcal{F}_t), \quad t \in [0,T].$$

Thus, $p_t \coloneqq (p_t^k)_{k\in[K]}$ is a random measure on $[K]$ representing the posterior (or filtered) (conditional) distribution, under $\mathbf{P}$, of the Markov chain Θ_t at time t:

$$\widehat{\Theta}_t = \sum_{k\in[K]} p_t^k e_k = p_t.$$

The filtered growth rate at time t is then simply the weighted average of the individual state growth rate under this measure:

$$\widehat{\gamma}^{(\Theta_t)} = \sum_{k=1}^{K} \gamma^{(k)} p_t^k.$$

If we are to have any hope of implementing the optimal portfolio 4.4 in practice, we need to understand the dynamics of the filter $\widehat{p}_t$. As a first step, we have the following celebrated result in filtering theory.

Theorem 5.4 (Fujisaki-Kallianpur-Kunita (FKK) Equation)**.** *The process p satisfies the SDE*

$$dp_t^k = p_t^\top G e_k dt + p_t^k \xi^{-1}\gamma(e_k - p_t)dN_t, \quad 1 \le k \le K, t \in [0,T], \tag{5.5}$$

where the process $N = (N_t)_{t\in[0,T]}$ is the innovations process characterized by the SDE

$$dN_t = \xi^{-1}\gamma(\Theta_t - \widehat{\Theta}_t)dt + dW_t, \quad t \in [0,T].$$

Proof. As the name suggests, Equation 5.5 is a direct application of the Fujisaki-Kallianpur-Kunita Theorem, as stated in [RW00] Theorem 8.11 for example. Letting $f \in C^0(E_K)$ be the indicator $f = \mathbf{1}_{e_k}$, it is clear that $f(\Theta_t) = \mathbf{1}_{\{\Theta_t = e_k\}}$ is in $\mathbf{L}^2$. The generator $\mathcal{G} : B(E_K) \to B(E_K)$ of the Markov Chain Θ_t is G acting by matrix multiplication, and hence we are considering the martingale

$$M_t^k \coloneqq \mathbf{1}_{\{\Theta_t = e_k\}} - \mathbf{1}_{\{\Theta_0 = e_k\}} - \int_0^t \Theta_s^\top G e_k \, ds.$$

Note that for each $1 \le i \le n$, the quadratic covariation process $\langle M^k, W^i\rangle$ vanishes, as M^k is the sum of a pure jump process and a finite variation process, and W^i is a continuous process. Consequently, the process

$$\alpha \coloneqq \frac{d}{dt}\langle M^k, W\rangle$$

vanishes. Thus, the FKK Equation given by Equation 8.16 of [RW00] is given by

$$dp_t^k = p_t^\top G e_k dt + p_t^k \xi^{-1}(\gamma^{(k)} - \gamma p_t)dN_t.$$

$\square$

The FKK Equation 5.5 is a nonlinear SDE governing the dynamics of the filter p. The key insight which allows us to obtain a more tractable representation for the filter is the existence of an equivalent measure $\widetilde{\mathbf{P}} \sim \mathbf{P}$, under which our observation process $\log X$ and our signal process Θ are independent.

We define the "market price of risk" process $\lambda = (\lambda_t)_{t\in[0,T]}$

$$\lambda_t \coloneqq \xi^{-1}\gamma^{(\Theta_t)}, \quad t \in [0,T], \tag{5.6}$$

which, like $\gamma^{(\Theta_t)}$, is unobservable, as well as its filtered counterpart:

$$\widehat{\lambda}_t \coloneqq \mathbf{E}[\lambda_t|\mathcal{F}_t] = \xi^{-1}\widehat{\gamma}^{(\Theta_t)} = \xi^{-1}\gamma\widehat{\Theta}_t, \quad t \in [0,T].$$

Lemma 5.7. *The process λ satisfies Novikov's condition:*

$$\mathbf{E}\left[\exp\left(\frac{1}{2}\int_0^T |\lambda_t|^2\,dt\right)\right] < \infty.$$

In particular, the process $Z = (Z_t)_{t\in[0,T]}$ defined by

$$Z_t \coloneqq \exp\left(-\int_0^t \lambda_s\,dW_s - \frac{1}{2}\int_0^t |\lambda_s|^2\,ds\right), \quad t \in [0,T]$$

is a $(\mathbb{G},\mathbf{P})$-martingale. We define the **probability of reference** $\widetilde{\mathbf{P}} \sim \mathbf{P}$ by the Radon-Nikodym derivative

$$\frac{d\widetilde{\mathbf{P}}}{d\mathbf{P}} \coloneqq Z_T. \tag{5.8}$$

It follows by Girsanov's theorem that the process $\widetilde{W} = (\widetilde{W}_t)_{t\in[0,T]}$ defined by

$$\widetilde{W}_t \coloneqq W_t + \int_0^t \lambda_s\,ds, \quad t \in [0,T]$$

is a $(\mathbb{G},\widetilde{\mathbf{P}})$-Brownian motion. The measure $\widetilde{\mathbf{P}}$ is a risk-neutral martingale measure for the market model 5.1, whose dynamics are given by

$$\begin{aligned}
d\log X_t &= \gamma^{(\Theta_t)}dt + \xi dW_t \\
&= \gamma^{(\Theta_t)}dt + \xi(d\widetilde{W}_t - \lambda_t dt) \\
&= \gamma^{(\Theta_t)}dt + \xi(d\widetilde{W}_t - \xi^{-1}\gamma^{(\Theta_t)}dt) \\
&= \xi d\widetilde{W}_t.
\end{aligned}$$

In particular, we see the observable σ-algebra $\mathbf{F}$ coincides with the augmentation of the canonical σ-algebra $\mathbf{F}^{\widetilde{W}}$ of the Brownian motion $\widetilde{W}$ under $\widetilde{\mathbf{P}}$. Furthermore, under the reference probability $\widetilde{\mathbf{P}}$, our observation process $Y = \xi^{-1}\log X$ is an n-dimensional Brownian motion.

In order to define a new filter distribution under $\widetilde{\mathbf{P}}$, we first define the process $Z^{-1} \coloneqq (\widehat{Z}_t)_{t\in[0,T]}$ by

$$Z_t^{-1} \coloneqq \frac{1}{Z_t}, \quad t \in [0,T],.$$

We note $Z_T^{-1} = d\mathbf{P}/d\widetilde{\mathbf{P}}$ and furthermore, given $t \in [0,T]$,

$$\begin{aligned}
Z_t^{-1} &= \exp\left(\int_0^t \lambda_s^\top\,dW_s + \frac{1}{2}\int_0^t |\lambda_s|^2\,ds\right) \\
&= \exp\left(\int_0^t \lambda_s^\top\,(d\widetilde{W}_s + \lambda_s ds) + \frac{1}{2}\int_0^t |\lambda_s|^2\,ds\right) \\
&= \exp\left(\int_0^t \lambda_s^\top\,d\widetilde{W}_s - \frac{1}{2}\int_0^t |\lambda_s|^2\,ds\right).
\end{aligned}$$

i.e., Z^{-1} is characterized by the SDE

$$\begin{cases} dZ_t^{-1} = Z_t^{-1}\lambda_t^\top d\widetilde{W}_t, \quad t \in [0,T] \\ Z_0^{-1} = 1 \end{cases}. \tag{5.9}$$

We extend the definition of the probability of reference 5.8 to a family $\{\widetilde{\mathbf{P}}_t\}_{t\in[0,T]}$ of $\mathbf{P}$-absolutely continuous measures, where

$$\frac{d\widetilde{\mathbf{P}}_t}{d\mathbf{P}} := Z_t$$

and their inverses

$$\frac{d\mathbf{P}_t}{d\widetilde{\mathbf{P}}_t} =: Z_t^{-1}.$$

Since Z (Z^{-1}) is a $(\mathbb{G}, \mathbf{P})$-martingale (resp. $(\mathbb{G}, \widetilde{\mathbf{P}})$-martingale), it follows that $\mathbf{P}_s$ and $\mathbf{P}_t$ (resp. $\widetilde{\mathbf{P}}_s$ and $\widetilde{\mathbf{P}}_t$) agree on $\mathbb{G}_s$ for $0 \le s \le t \le T$. Given $t \in [0,T]$, we let $\mathbf{E}_t$ and $\widetilde{\mathbf{E}}_t$ denote the expectations under $\mathbf{P}_t$ and $\widetilde{\mathbf{P}}_t$ respectively. We suppress the subscript in the case $t = T$.

Theorem 5.10 (Kallianpur-Striebel (KS) Formula)**.** *We have the Bayes-like formula*

$$\widehat{\Theta}_t = \frac{\widetilde{\mathbf{E}}[Z_t^{-1}\Theta_t|\mathcal{F}_t]}{\widetilde{\mathbf{E}}[Z_t^{-1}|\mathcal{F}_t]} \tag{5.11}$$

for all $t \in [0,T]$.

Proof. Given an event $A \in \mathcal{F}_t$, we have

$$\begin{aligned}
\mathbf{E}\left[\mathbf{1}_A \frac{\widetilde{\mathbf{E}}[Z_t^{-1}\Theta_t|\mathcal{F}_t]}{\widetilde{\mathbf{E}}[Z_t^{-1}|\mathcal{F}_t]}\right] &= \mathbf{E}_t\left[\mathbf{1}_A \frac{\widetilde{\mathbf{E}}[Z_t^{-1}\Theta_t|\mathcal{F}_t]}{\widetilde{\mathbf{E}}[Z_t^{-1}|\mathcal{F}_t]}\right] \\
&= \widetilde{\mathbf{E}}\left[Z_t^{-1}\mathbf{1}_A \frac{\widetilde{\mathbf{E}}[Z_t^{-1}\Theta_t|\mathcal{F}_t]}{\widetilde{\mathbf{E}}[Z_T^{-1}|\mathcal{F}_t]}\right] \\
&= \widetilde{\mathbf{E}}\left[\widetilde{\mathbf{E}}[Z_t^{-1}|\mathcal{F}_t]\mathbf{1}_A \frac{\widetilde{\mathbf{E}}[Z_t^{-1}\Theta_t|\mathcal{F}_t]}{\widetilde{\mathbf{E}}[Z_t^{-1}|\mathcal{F}_t]}\right] \\
&= \widetilde{\mathbf{E}}[\mathbf{1}_A Z_t^{-1}\Theta_t] \\
&= \mathbf{E}[\mathbf{1}_A \Theta_t]
\end{aligned}$$

where the third equality follows the definition of conditional expectation and the $\mathcal{F}_t$-measurability of the integrand (modulo Z_t^{-1}). We conclude the right-hand side of Equation 5.11 satisfies the defining property of the filter as a conditional expectation. ☐

Definition 5.12. We call the conditional expectation $\widetilde{\mathbf{E}}[Z_t^{-1}\Theta_t|\mathcal{F}_t]$ appearing in Equation 5.11 the **unnormalized filter** of the latent state Θ, and we denote it by ρ_t. Furthermore, given $k \in [K]$, we denote by ρ_t^k the conditional expectation $\widetilde{\mathbf{E}}[Z_t^{-1}\mathbf{1}_{\{\Theta_t = e_k\}}|\mathcal{F}_t]$, so that

$$\rho_t = \sum_{k=1}^K \rho_t^k e_k$$

In particular, we see that the Kallianpur-Striebel Formula 5.10 allows us to express the filter distribution p_t in terms of the unnormalized filter ρ_t:

$$p_t = \sum_{k=1}^K p_t^k e_k = \widehat{\Theta}_t = \frac{\rho_t}{\sum\limits_{k=1}^K \rho_t^k}. \tag{5.13}$$

Equation 5.13 justifies the terminology in Definition 5.12 of ρ as being *unnormalized.* To further this analogy, we write $\rho(\mathbf{1}) = (\rho_t(\mathbf{1}))_{t\in[0,T]}$ for the observable process

$$\rho_t(\mathbf{1}) \coloneqq \sum_{k=1}^{K} \rho_t^k = \widetilde{\mathbf{E}}[Z_t^{-1}|\mathcal{F}_t], \quad t \in [0,T]. \tag{5.14}$$

The advantage of passing from the filter p to ρ is that the latter is characterized by a *linear* observable SDE known as the *Zakai equation.* Before stating and proving the Zakai equation, we need a preliminary result concerning filters of stochastic integrals.

Lemma 5.15 (Brownian Fubini)**.**

1. *Suppose $\lambda \in \mathbf{L}^2$. Then*
$$\mathbf{E}\left[\int_0^t \lambda_s\, dW_s \middle| \mathcal{F}_t^W\right] = \int_0^t \mathbf{E}[\lambda_s|\mathcal{F}_t^W]\, dW_s$$
for all $t \in [0,T]$
2. *Suppose M is a semimartingale independent of W and $\lambda \in L^2(\langle M\rangle)$. Then*
$$\mathbf{E}\left[\int_0^t \lambda_s\, dM_s \middle| \mathcal{F}_t^W\right] = 0$$
for all $t \in [0,T]$.

Proof. We begin with the first statement. Given $A \in \mathcal{F}_t^W$, by the Martingale Representation Theorem, there exists $\xi \in \mathbf{L}^2$, $\mathbf{F}^W$-adapted, such that

$$\mathbf{1}_A = \mathbf{P}(A) + \int_0^t \xi_t\, dW_t.$$

In particular, by the Itô isometry, we have

$$\begin{aligned}
\mathbf{E}\left[\mathbf{1}_A \int_0^t \lambda_s\, dW_s\right] &= \mathbf{E}\left[\int_0^t \xi_s\, dW_s \int_0^t \lambda_s\, dW_s\right] \\
&= \mathbf{E}\left[\int_0^t \xi_s \lambda_s\, ds\right] \\
&= \mathbf{E}\left[\int_0^t \xi_s \mathbf{E}[\lambda_s|\mathcal{F}_t^W]\, ds\right] \\
&= \mathbf{E}\left[\int_0^t \xi_s\, dW_s \int_0^t \mathbf{E}[\lambda_s|\mathcal{F}_t^W]\, dW_s\right] \\
&= \mathbf{E}\left[\mathbf{1}_A \int_0^t \mathbf{E}[\lambda_s|\mathcal{F}_t^W]\, dW_s\right]
\end{aligned}$$

where the third equality follows from conditional Fubini's theorem.

For the second statement, we repeat the above proof, except noting that the Itô isometry now yields

$$\begin{aligned}
\mathbf{E}\left[\mathbf{1}_A \int_0^t \lambda_s\, ds\right] &= \mathbf{E}\left[\int_0^t \xi_s \lambda_s\, d\langle M, W\rangle_t\right] \\
&= 0
\end{aligned}$$

where the Lebesgue-Stieltjes integral vanishes due to the fact that independence implies the covariation $\langle M, W\rangle = 0$ vanishes.

□

Theorem 5.16 (Zakai Equation)**.** *The unnormalized filter process ρ satisfies the SDE*

$$\begin{cases} d\rho_t = G^\top \rho_t dt + \operatorname{diag}(\rho_t)\gamma^\top \Sigma^{-1} d\log X_t, & t \in [0,T] \\ \rho_0 = p_0 = \pi \end{cases}.$$

Proof. The standard approach for deriving the Zakai equation is applying Itô's product rule to the pair $\mathbf{1}_{\{\Theta_t = e_k\}} Z_t^{-1}$, and then utilizing the Brownian Fubini Theorem to determine the differential of the corresponding filtered process. We have

$$d(\mathbf{1}_{\{\Theta_t=e_k\}} Z_t^{-1}) = Z_t^{-1} d\mathbf{1}_{\{\Theta_t=e_k\}} + \mathbf{1}_{\{\Theta_t=e_k\}} dZ_t^{-1} + \underbrace{d\langle \mathbf{1}_{\{\Theta=e_k\}} Z^{-1}\rangle_t}_{=0} \tag{5.17}$$

where the differential of the covariation vanishes because $\mathbf{1}_{\{\Theta=e_k\}}$ is a pure jump process and Z^{-1} is a continuous process, as indicated by the SDE 5.9. Recalling the notation from the proof of FKK Equation 5.4, note

$$d\mathbf{1}_{\{\Theta_t=e_k\}} = \Theta_t^\top G e_k dt + dM_t^k,$$

where M^k is an $\mathbf{F}^\Theta$-adapted $\mathbf{L}^2$ (in fact $\mathbf{L}^\infty$) martingale with $M_0^k = 0$. Thus 5.17 becomes

$$d(\mathbf{1}_{\{\Theta_t=e_k\}} Z_t^{-1}) = Z_t^{-1}\Theta_t^\top G e_k dt + Z_t^{-1} dM_t^k + \mathbf{1}_{\{\Theta_t=e_k\}} Z_t^{-1} (\xi^{-1}\gamma^{(k)})^\top d\widetilde{W}_t. \tag{5.18}$$

Since all coefficients in Equation 5.18 are in $\mathbf{L}^2$, exchanging integration and conditional expectation in the first summand and applying the Brownian Fubini Theorem 5.15 parts (2) and (1) to the other two, 5.18 becomes

$$d\rho_t^k = \rho_t^\top G e_k dt + \rho_t^k (\gamma^{(k)})^\top (\xi^{-1})^\top d\widetilde{W}_t.$$

Finally, the proof is finished by recalling $\xi^{-1} d\log X_t = d\widetilde{W}_t$. □

Note combining the Zakai Equation 5.16 together with the KS Formula 5.10 yield an alternative derivation of the FKK equation 5.4.

With the Zakai equation giving us a means of simulating the filter p, all that remains is to formulate the optimal portfolio under the hidden Markov model.

Corollary 5.19 (Optimal portfolio for HMM)**.** *The optimal portfolio $\pi^{\mathrm{HMM}} = (\pi_t^{\mathrm{HMM}})_{t\in[0,T]}$ solving the SCP 4.3 in the case of the market model 5.1 is given by*

$$\pi_t^{\mathrm{HMM}} := \sum_{k=1}^K p_t^k \pi_t^{(k)}$$

where $\pi_t^{(k)}$ is the solution to the SCP conditioned on $\{\Theta_t = k, \forall t \in [0,T]\}$, i.e., under the market model

$$d\log X_t = \gamma^{(k)} dt + \xi dW_t, \quad t \in [0,T]. \tag{5.20}$$

Proof. Recall the processes A^{HMM} and B^{HMM} defined by Equation 4.14 and used in the definition of the optimal portfolio 4.4. Using the fact that the filtered growth rate of the HMM is given by $\widehat{\gamma} = \gamma p_t = \sum_{k=1}^n p_t^k \gamma^{(k)}$, and $\mathbf{1}^\top p \equiv 1$,

$$\begin{aligned} B_t^{\mathrm{HMM}} &= \sum_{k=1}^K \xi^0 p_t^k \gamma^{(k)} + \frac{1}{2}\operatorname{diag}(\widehat{\Sigma}_t) - \xi^1 \Omega_t \eta_t \\ &= \sum_{k=1}^K p_t^k (\xi^0 \gamma^{(k)} + \frac{1}{2}\operatorname{diag}(\widehat{\Sigma}_t) - \xi^1 \Omega \eta) \\ &= \sum_{k=1}^K p_t^k B_t^k \end{aligned}$$

where B^k is the corresponding process for the market model 5.20. Since the optimal portfolio is linear in B and $A^{\mathrm{HMM}} \equiv A^k$ for all $1 \le k \le K$, the result follows. □

6 Implementation of the optimal portfolio under the HMM

Following [AAJ21], we simulate the optimal portfolio 5.19 solving the SCP under the hidden Markov market model 5.1 using real-world historic stock price data. The task of carrying out such an experiment consists of roughly the following steps:

(1) Determine the model dataset. This includes choosing a market universe $X = (X^1, \ldots, X^n)$ of equities as well as the historical dataset to use for our backtests.

(2) Discretize the Zakai equation.

(3) Calibrate the filtered state variable $\widehat{\Theta}$, the filtered market growth rates $\widehat{\gamma}$ and the covariation matrix $\widehat{\Sigma}$ on historic (in-sample) data.

(4) Estimate the preference parameters $\xi = (\xi^0, \xi^1_t, \xi^2_t)$ on historic (in-sample) data.

(5) Carry out rolling (out-of-sample) backtests using estimated parameters from (3) and (4). Evaluate relative performance of optimal portfolio π^* against preselected benchmark η.

For a given backtest $b \in [B]$ beginning at time t_0, we consider a finite set $\mathcal{T} \coloneqq \{t_n\}_{n=0}^{N-1}$ of N discrete time points, where $t_n \coloneqq t_0 + n\Delta t$.

6.1 The data

In [AAJ21], Al-Aradi and Jaimungal use as their market universe 12 industry portfolios constructed from stocks listed on the NYSE, AMEX, and NASDAQ according to their Standard Industrial Classification (SIC) codes. The industries consist of: Consumer Non-durables, Consumer Durables, Manufacturing, Energy, Chemicals, Telecommunications, Utilities, Shops, Health, Money, and Other). The rationale behind their choice of market universe is as follows:

1. The size ($n = 12$) of their universe was chosen so as to circumvent convergence difficulties which would likely arise in estimating the parameters of the HMM.
2. By using industry portfolios in place of individual stocks, we do not need to worry about corporate events which could bias our dataset (e.g. M&A, bankruptcies, etc).

They use a dataset consisting of daily percentage returns for those portfolios acquired from Kenneth French's well-known data library.[1] Their backtest observation period is January 1970 to December 2017. We utilize the same dataset for our experiment, but we extend the observation period until December 2024.

6.2 Discretizing the Zakai equation

In order to implement the optimal portfolio 5.19, we need to compute the filter distribution process p_t, which by the Kallianpur-Striebel formula 5.10, we know reduces to estimating, and hence discretizing, the Zakai equation 5.16. The method Al-Aradi and Jaimungal apply is slightly more involved than simply applying the classical Euler-Maruyama discretization scheme, as the latter suffers from instability leading to underflow. As we work with daily returns data and we assume time is measured in years, we take $\Delta t = \frac{1}{252}$ as our discretization time step.

We first approximate the generator matrix G of the latent continuous-time Markov state Θ with the transition matrix $Z \coloneqq e^{\Delta t G}$ for the discrete-time Markov chain $\{\Theta_n\}_{n=0}^{N-1}$.

Motivated by the exponential form of the time differential in 5.16, we posit that for each $0 \leq n < N$, the unnormalized filter satisfies the recursive relation

$$\rho_t = e^{(t-t_n)G^\top} \rho_{t_n} * Y_t^{(n)}, \quad t \in [t_n, t_{n+1}], \tag{6.1}$$

[1] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

where $*$ denotes the Hadamard product, and $Y^{(n)} = (Y_t^{(n)})_{t\in[t_n,t_{n+1}]}$ is a process initialized at $Y_{t_n}^{(n)} = 1$. Consequently, by Itô's formula, we have

$$d\rho_t = G^\top \rho_t + e^{(t-t_n)G}\rho_u * dY_t^{(n)}, \quad t \in [t_n, t_{n+1}].$$

Equating this with the Zakai equation yields

$$\rho_u * dY_t^{(n)} = e^{-(t-t_n)G^\top}\operatorname{diag}(\rho_t)\gamma^\top \Sigma^{-1} d\log X_t.$$

Finally, dividing out the unnormalized filter entrywise, we get

$$\begin{aligned} dY_t^{(n)} &= e^{-(t-t_n)G^\top}\operatorname{diag}(\rho_t/\rho_{t_n})\gamma^\top \Sigma^{-1} d\log X_t \\ &= \operatorname{diag}(Y_t)\gamma^\top \Sigma^{-1} d\log X_t. \end{aligned}$$

We see that $Y^{(n)}$ has no dependence on the interval $[t_n, t_{n+1}]$, hence we drop the superscript (n). Integrating over $[t_n, t_{n+1}]$ yields

$$Y_t = \exp\left(\gamma^\top \Sigma^{-1}\log(X_t/X_{t_n}) - \frac{1}{2}\gamma^\top\Sigma^{-1}\gamma(t-t_n)\right), \quad t \in [t_n, t_{n+1}].$$

Subsisting this expression into Equation6.1, we arrive at the recursive relation characterizing the discretized unnormalized filter:

$$\rho_{t_{n+1}} = Z^\top \rho_{t_n}\exp\left(\gamma^\top\Sigma^{-1}\log(X_{n+1}/X_n) - \frac{1}{2}\gamma^\top\Sigma^{-1}\gamma\Delta t\right), \quad 0 \le n < N. \tag{6.2}$$

6.3 HMM calibration schemes

Our HMM consists of the following parameters $\theta \coloneqq (\pi, Z, \gamma, \Sigma)$ to be estimated:

Hidden state process (Θ_t)	Observation process $(\log X_t)$
• initial distribution $\pi \in \mathcal{P}([K])$	• market growth rates $\gamma \in \mathbf{R}^{n\times k}$
• transition matrix $Z \in \mathbf{R}^{2k}$	• covariance matrix $\Sigma \in \mathrm{Sym}_{>0}(\mathbf{R}^n)$

Given the return data available to us as described in section 6.1 and fixing a backtest $b \in [B]$, the particular model we apply the EM algorithm to is a discrete-time HMM $\boldsymbol{R}_N = \{r_n\}_{n=0}^{N-1}$ given by the daily logarithm returns of our market at the time points $t_n = t_0 + n\Delta t$. Given the Itô dynamics 5.1 of our market model, we have

$$r_n \coloneqq \log X_{t_n+\Delta t} - \log X_{t_n} \sim N\left(\int_{t_n}^{t_n+\Delta t}\gamma\Theta_s\, ds, \Delta t\Sigma\right), \quad 0 \le n \le N-1.$$

Under the assumption that the latent Markov chain Θ only switches states at times in $\mathcal{T} = \{t_n\}_{n=0}^{N-1}$, the distribution of the log returns simplifies to

$$r_n \sim N(\gamma\Theta_n, \Delta t\Sigma), \quad 0 \le n < N. \tag{6.3}$$

The discrete-time process $\boldsymbol{R} = \{r_n\}_{n=0}^{N-1}$ is the HMM which we estimate.

6.3.1 The EM approach

The most widely used method of estimating the parameters of a hidden Markov model is the *Expectation Maximization (EM) algorithm*[2] outlined in Appendix B. The following three figures depict various model estimates for the HMM trained via the EM algorithm on 5 years of daily return data beginning January 2020.

[2]The EM algorithm is also often referred to as the Baum-Welch algorithm, the latter in fact describing the forward-backward algorithm used to implement EM in the case of HMMs.

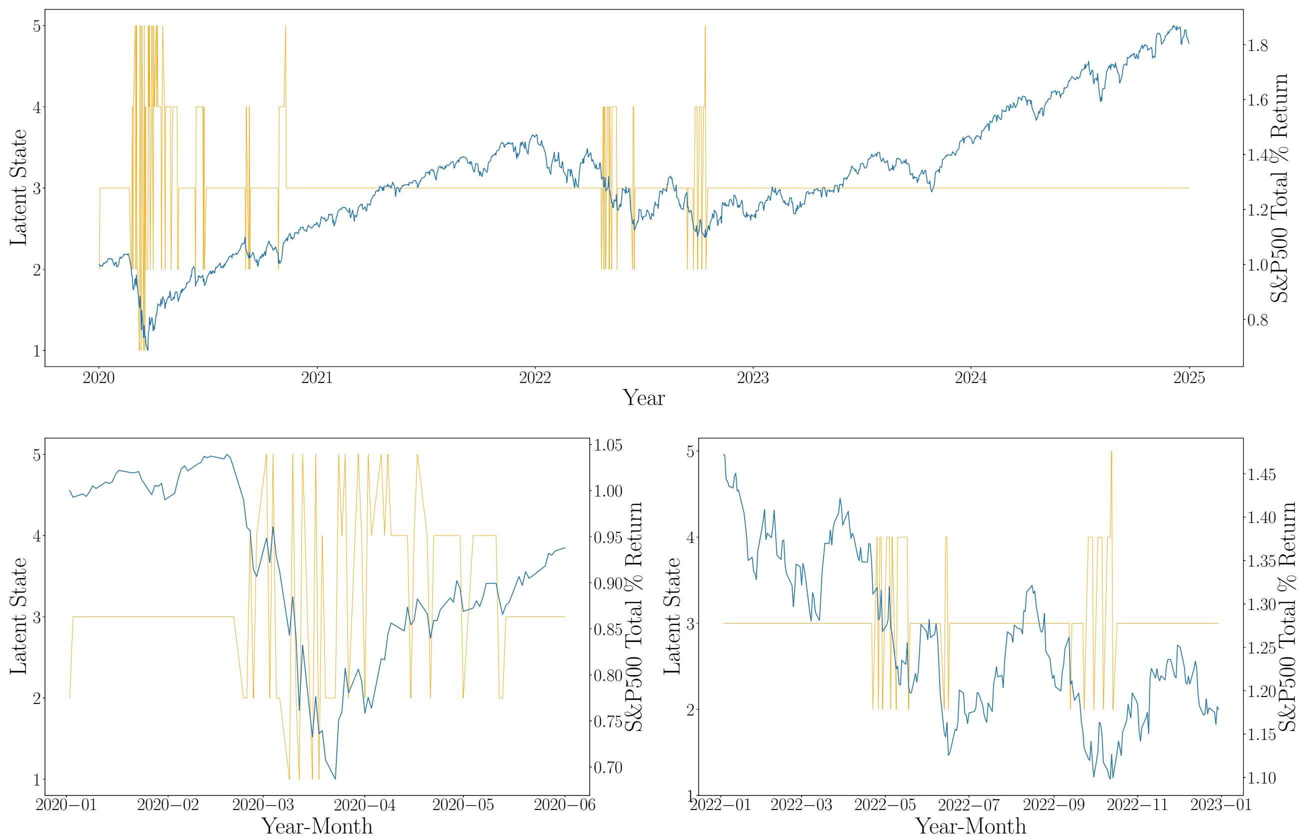

Figure 2: Top panel: most likely trajectory of latent Markov state Θ (in orange) over 5-year period beginning January, 2020 according to the Viterbi algorithm, overlaid with S&P500 total return (in blue). Bottom left and right panels: Close ups of Viterbi path during periods of high intensity regime-switching (2020 COVID-19 crash and 2022 bear market respectively).

Figure 3: Hidden Markov market model parameter estimation for $K = 5$ latent states and 5-year estimation window beginning January, 2020. Top panel: daily return vectors derived from estimated daily growth rates $\Delta t \gamma^{(k)}$ for each state $k \in \{1, \ldots, 5\}$. Stats are labelled by increasing average daily growth rate across industries. Bottom panel: annualzed volatilities (left) and correlations for industry portfolios estimated from covariance matrix Σ.

State	Average Daily Growth Rate Across Industries	Number of Days Spent in State	% of Days Spent in State	Expected Sojourn Time
State 1 (Very Bad)	-914 bps	4	0.3%	1 day
State 2 (Bad)	-297 bps	37	2.9%	1 day
State 3 (Normal)	12 bps	1134	90.1%	94 days
State 4 (Good)	46 bps	69	5.5%	3 days
State 5 (Very Good)	492 bps	14	1.1%	1 day

Table 1: HMM emissions $\boldsymbol{R}$ estimates from applying the EM algorithm to the 5-year estimation window beginning January, 2020. Estimate of days spent in each state based on MAP estimate of $\{\Theta_n\}_{n=0}^N$ via Viterbi algorithm.

6.3.2 Order selection

In addition to the parameters listed in table 6.3, we also have a hyperparameter given by the number K of hidden states of the latent Markov chain Θ to determine. We say the HMM $(\log X, \Theta)$ has **order** K if Θ is a K state Markov chain, which we emphasize by writing $\Theta(K)$. We begin by noticing that a $K+1$ state Markov chain $\Theta(K+1)$ can always simulate the law of a K state Markov chain $\Theta(K)$, by defining $\Theta(K+1)$ as the pushforward along $[K] \hookrightarrow [K+1]$. In particular, the law of the corresponding observation process $\log X$ under $\Theta(K+1)$ and $\Theta(K)$ coincide. Consequently, as a function of the order, the likelihood $\mathcal{L}^\theta(K) \coloneqq p(r_0, \ldots, r_{N-1}|\theta)$ of the observation sample $\boldsymbol{R}$ is nondecreasing. To accurately estimate the true order K^* of the sample $\{(r_n, \Theta_n)\}_{n=0}^{N-1}$, we employ four different methods, outlined below. The first three are penalized maximum likelihood estimators of the form

$$\widehat{K} = \underset{K>0}{\operatorname{argmax}}\{\log \mathcal{L}^*(K) - P(N,K)\}$$

where $\mathcal{L}^*(K)$ denotes the likelihood of the observation sample $\boldsymbol{R}_N$ under the EM parameter set θ^*, and $P(T,K)$ is a regularization term meant to penalize the use of erroneous latent states. The last order criterion we employ is a cross-validated likelihood meant to estimate the out-of-sample performance of the HMM model for a given choice of order K.

1. (*Akaike information criteria*). The penalty term $P_{\mathrm{AIC}}(N,K)$ for the Akaike information criterion (AIC) is given by
$$P_{\mathrm{AIC}} = |\theta|$$
where $|\theta|$ denotes the number of parameters to be estimated in our model. In the case of an order K HMM, we have
$$|\theta(K)| = K + K(K-1) + nK + \frac{K(K+1)}{2},$$
with the summands corresponding to the parameters of π, Z, γ, Σ respectively.

2. (*Bayesian information criteria*). The penalty term $P_{\mathrm{BIC}}(N,K)$ for the Bayesian information criterion (BIC) is
$$P_{\mathrm{BIC}}(N,K) = \frac{1}{2}|\theta|\ln(N).$$

3. (*Integrated completed likelihood*). The Integrated completed likelihood (ICL) is a penalized maximum likelihood estimator of the form
$$\mathrm{ICL}(N,K) \coloneqq \log p(r_0, \ldots, r_{N-1}|\boldsymbol{\Theta}^{\mathrm{MAP}}) - \frac{1}{2}|\theta|\ln(N),$$
where:

- $\mathbf{\Theta}^{\text{MAP}} := (\widehat{\Theta}_n)_{n=0}^{N}$ is the maximum a posteriori (MAP) estimate for the latent state path:

$$\mathbf{\Theta}^{\text{MAP}} := \underset{\mathbf{\Theta} \in E_K^N}{\operatorname{argmax}}\, p(r_0, \ldots, r_{N-1} | \mathbf{\Theta}),$$

computed via the Viterbi algorithm ([Bis06], 13.2.5);

- the regularization penalty $P_{\text{ICL}}(N, K) = P_{\text{BIC}}(N, K)$ coincides with that of the BIC.

4. (*Odd-even half sampling*). Formulated in [CM08], Odd-even half sampling is a two-fold cross-validation likelihood estimator based on the following observation. Letting $N' := \lfloor (N-1)/2 \rfloor$ and $N'' := \lceil (N-1)/2 \rceil$, if the (π, Z)-Markov chain $(\Theta_n)_{n=0}^{N-1}$ is assumed stationary, then

$$\mathbf{\Theta}^E := \{\Theta_{t_{2n}}\}_{n=0}^{N'} \quad \text{and} \quad \mathbf{\Theta}^O := (\Theta_{n_{2i+1}})_{n=0}^{N'-1}$$

are both (π, Z)-Markov chains. Moreover, $\boldsymbol{R}^E := \{r_{2n}\}_{n=0}^{N'}$ and $\boldsymbol{R}^O := \{r_{2n+1}\}_{n=0}^{N''}$ are HMMs with the same parameter set, namely, the observations r_n in either case have Gaussian mixture distributions of the form 6.3, where $\mathbf{\Theta}$ is replaced with $\mathbf{\Theta}^E$ and $\mathbf{\Theta}^O$ respectively.

The OEHS estimate for a given model order K is then the average cross-validated log likelihood of the HMMs $\boldsymbol{R}^E$ and $\boldsymbol{R}^O$:

$$\text{OEHS}(K) := \frac{1}{2}(\log \mathcal{L}^{O,*} + \log \mathcal{L}^{E,*}),$$

where $\mathcal{L}^{O,*} = p(\boldsymbol{R}^O | \theta^{E,*})$ denotes the likelihood of the odd returns $\boldsymbol{R}^O$ with respect to the parameter set $\theta^{E,*}$ given by applying the EM algorithm to the even HMM $\boldsymbol{R}^E$, and vice versa for $\mathcal{L}^{E,*}$.

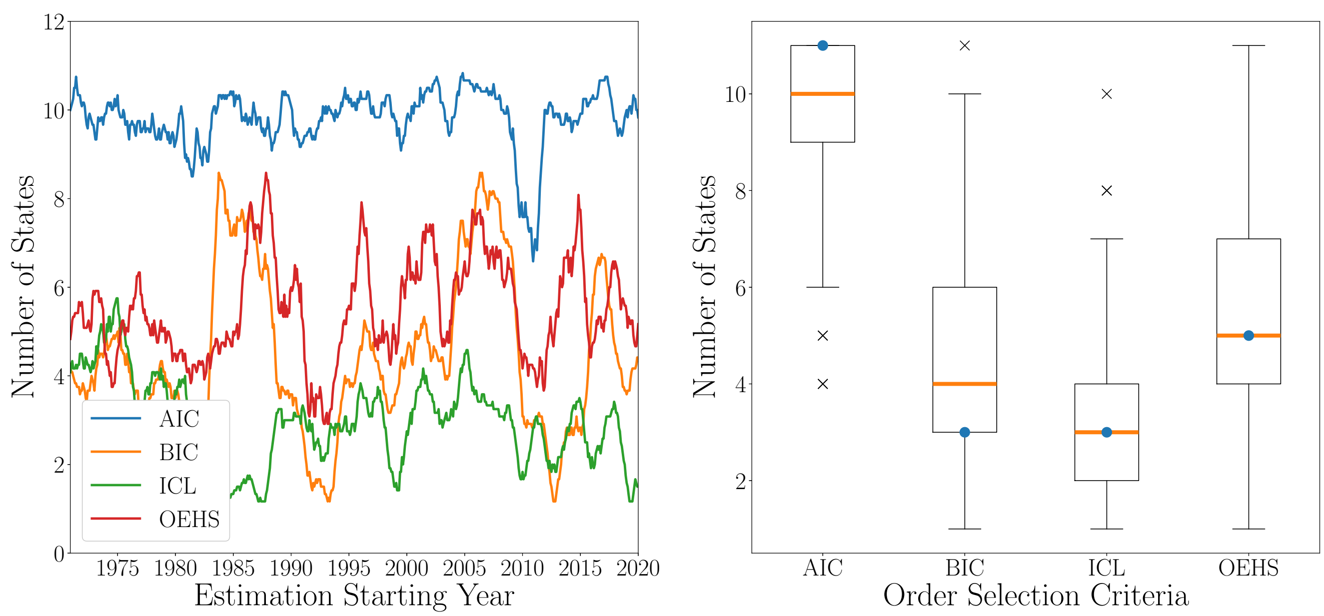

Figure 4: Left panel: Rolling 12-month average HMM order $\widehat{K}$ estimate under various order selection criteria and 5-year estimation window. Right panel: Box plot for order estimates under various order selection criteria. Mode and median order estimate denoted by blue dot and orange line respectively.

6.4 Investment parameter selection and estimation

The particular optimal portfolio π^{HMM} we simulate depends on the portfolio optimization problem specified by our choice of the parameter set $\{\rho, \Omega, Q, \xi\}$ for the performance criteria 4.3. We follow [AAJ21], taking as both our outperformance and tracking benchmark portfolios $\pi^{\mathrm{EW}} \in \mathcal{P}$, the equal-weight portfolio

$$\pi^{\mathrm{EW}} := \frac{1}{n}\mathbf{1}.$$

It is well established (e.g., [DGU07]) that the equal-weight portfolio generally outperforms the market (cap-weighted) portfolio, the latter being more commonly used in SPT. We fix the penalty covariance processes $\Omega = \Sigma = Q$, i.e., we penalize absolute risk as well as deviation from the equal-weight portfolio.

6.5 Backtest and performance evaluation

Our in-sample parameter estimation window size is 5 years, beginning January, 1970 and ending January, 2019. Our out-of-sample optimal portfolio backtest is then carried out over the subsequent year. We then roll over one month and repeat this estimation and evaluation procedure. In this way, we have a total of $B = 589$ separate backtests.

Following [AAJ21], in order to test the HMM model 5.1 for robustness against model misspecification, we evaluate the optimal portfolio under the various model order selection criteria outlined in the previous section.

Consider a backtest $b \in [B]$ consisting of N samples at time points $t_n = t_0 + n\Delta t$, $0 \leq n < N$. Let $T = t_0 + (N-1)\Delta t$ denote the terminal time. We use the following metrics to evaluate the risk-return performance of the optimal portfolio in the backtest b:

1. We measure outperformance of the optimal portfolio via the relative rate of return at terminal time:

$$\mathrm{OP}(b) := V_T^{\pi^{\mathrm{HMM}},\pi^{\mathrm{EW}}} - 1.$$

2. We measure the active return of the optimal portfolio via the annualized mean excess daily rate of return with respect to the benchmark portfolio:

$$\mathrm{ARet}(b) := \frac{1}{\Delta t}\frac{1}{N}\sum_{n=0}^{N-1}(\pi_n^{\mathrm{HMM}} - \pi_n^{\mathrm{EW}})^\top r_n(b).$$

3. We measure the active risk of the optimal portfolio via the annualized standard deviation of the excess daily rate of return with respect to the benchmark portfolio:

$$\mathrm{ARisk}(b) := \left(\frac{1}{\Delta t}\frac{1}{N}\sum_{n=0}^{N-1}((\pi_n^{\mathrm{HMM}} - \pi_n^{\mathrm{EW}})^\top r_n(b) - \Delta t\mathrm{ARet}(b))^2\right)^{\frac{1}{2}}.$$

4. Finally, we also consider the *gain-loss ratio* of the optimal portfolio across all of the backtests, defined by

$$\mathrm{GLR} := \frac{\mathbf{E}_{b\in[B]}[\mathrm{OP}(b)|\mathrm{OP}(b) > 0]}{\mathbf{E}_{b\in[B]}[\mathrm{OP}(b)|\mathrm{OP}(b) < 0]}.$$

6.6 Results and Conclusion

The performance results of our backtests are summarized by Figures 6.6 and 6.6. We begin our discussion of the backtest results by recalling the two-fold motivation of our numerical experiments:

(1) We would like assess the performance of the optimal portfolio 5.19 for the hidden Markov market model on new market data, namely the backtesting period between February 2017 and December 2024.

(2) We are attempting to reproduce the numerical results carried out in [AAJ21].[3]

[3] This of course relies on the (possibly unjustified) assumption that the present author has correctly implemented the code outlined in [AAJ21], so that the above results can be considered a faithful reproduction.

The two objectives are closely linked, as the performance metrics computed over all backtests $[B]$ consists of the 505 backtests $[b_{01/2017}]$ carried out in [AAJ21], as well as 84 new backtests corresponding to previously unseen market return data.

On the topic of robustness of the HMM to model misspecification, we see the performance metrics are relatively consistent across the various model selection criteria. In general, even when compared within the same observation window (1975-2017), our performance (both in terms of outperformance and risk) are considerably worse than those in [AAJ21]. This discrepancy most likely stems from the differences in our model order estimates, especially under the BIC and ICL selection criteria. In general, our order estimates were significantly less than those in [AAJ21], and under the selection criteria whose results were most similar (AIC), our corresponding backtests come closest to reproducing the results in [AAJ21]. It would be an interesting to learn what led to this difference in order estimation.

This expository work outlined the major theoretical results in Al-Aradi and Jaimungal's paper [AAJ21] on solving a stochastic control portfolio selection problem with latent model parameters and relative performance criteria. We also reproduced and extended the numerical experiments in [AAJ21], making the code publicly available for ease of future reproduction and extension. An interesting area of future research would be to produce and implement optimal portfolios under difference choices of convex performance criteria than 4.5.

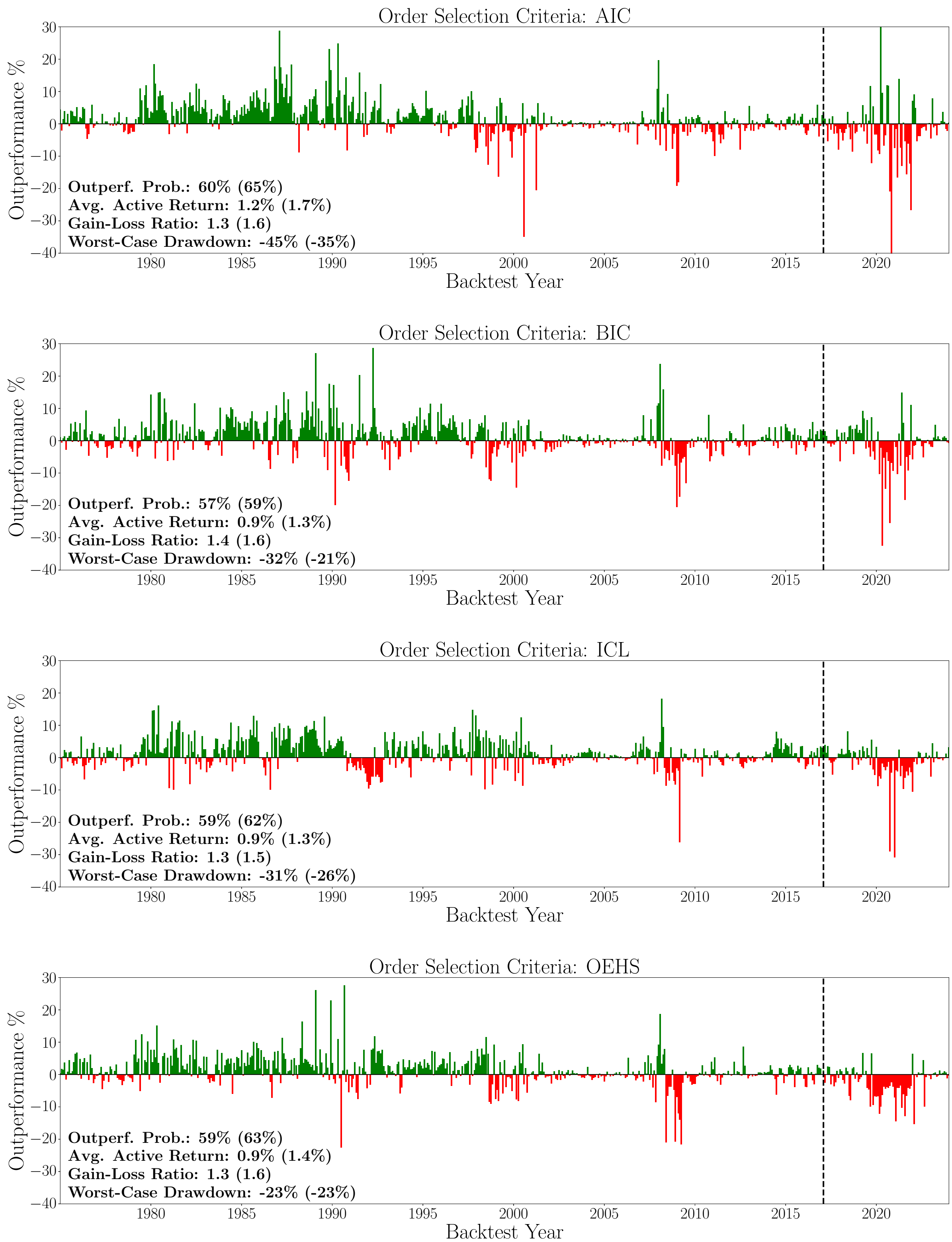

Figure 5: Active return for each backtest under various model order selection criteria. The dashed vertical line denotes the last backtest $b_{01/2017} = 505$ in [AAJ21]. Performance metrics evaluated up to $b_{01/2017}$ given in parenthesis.

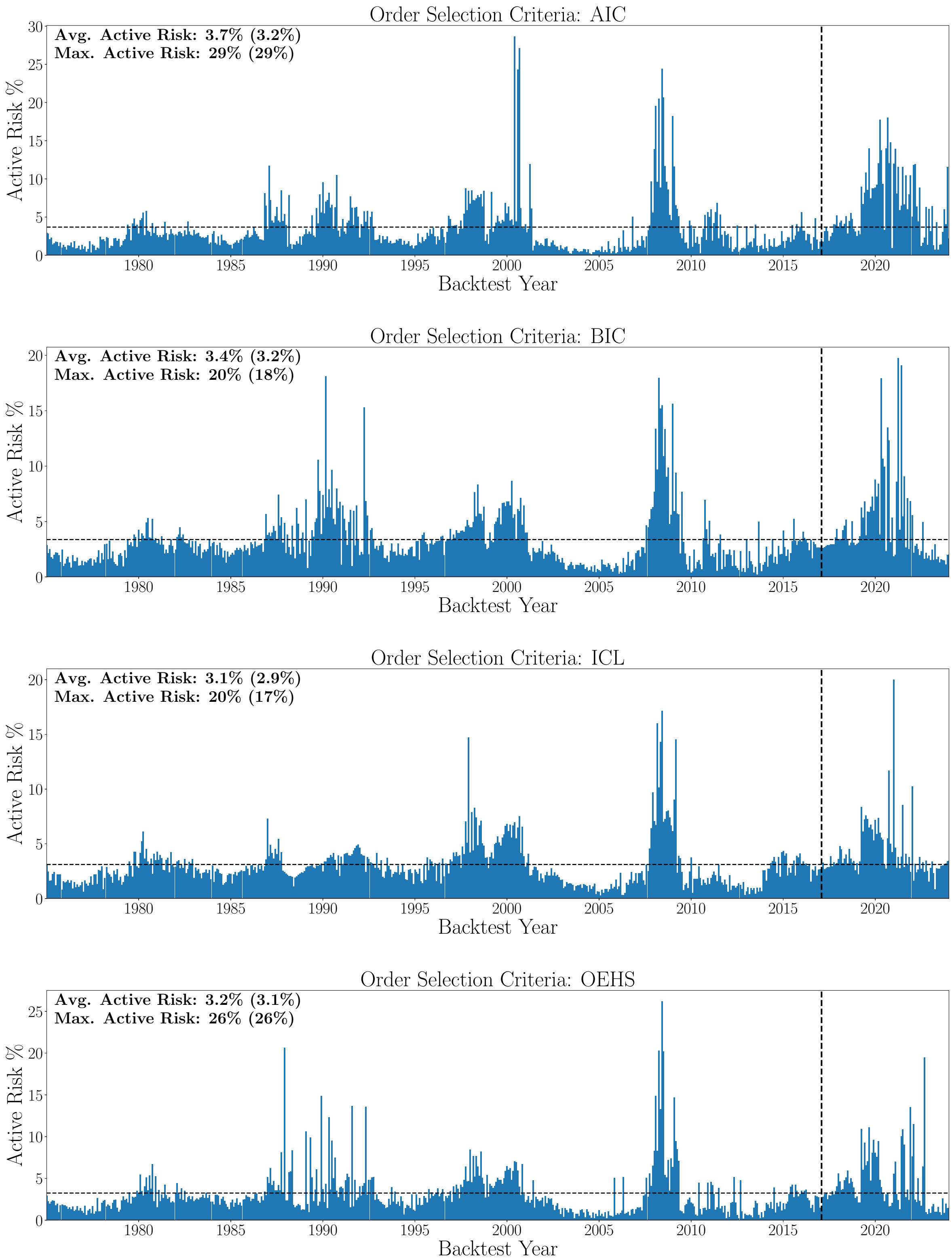

Figure 6: Active risk for each backtest under various model order selection criteria. The dashed horizontal line denotes average active risk and the dashed vertical line denotes the last backtest $b_{01/2017} = 505$ in [AAJ21]. Performance metrics evaluated up to $b_{01/2017}$ given in parenthesis.

Appendix A. The Gâteaux derivative

We give a brief recollection of some concepts in convex analysis following Chapters 1 and 2 of [ET99] which are needed for our proofs.

Let X denote an arbitrary Banach space, which we fix throughout this section.

Definition A.1. We say X is **reflexive** if its unit ball is compact in the weak topology.

Let $C \subset X$ be a nonempty closed convex subset, and consider a functional $F\colon C \to \overline{\mathbf{R}}$. We denote by $X^* := B(X, \mathbf{R})$ the dual space of bounded linear functionals on X, and by $\langle -, - \rangle$ the canonical pairing

$$\begin{aligned} \langle -, - \rangle \colon X \times X^* &\longrightarrow \mathbf{R} \\ (u, \varphi) &\longmapsto \varphi(u). \end{aligned}$$

Suppose we are interested in the optimization problem

$$\inf_{u \in C} F(u).$$

Definition A.2. An element $u^* \in C$ is a **solution of the optimization problem for** F if

$$u^* \in \operatorname*{argmin}_{u \in C} F(u).$$

We are interested in the study of optimization problems in which the functional F is convex, defined appropriately given the fact that F can attain the values $\pm\infty$.[4]

Definition A.3. The functional F is **convex** if

$$F(\lambda u + (1-\lambda)v) \leq \lambda F(u) + (1-\lambda)F(v)$$

for every $\lambda \in (0,1)$, and every $u, v \in C$ for which the right-hand side is well-defined, i.e., such that $F(u) = -F(v) = \pm\infty$ does not hold.

Remark A.4 (Extremal behaviour under convexity)**.** The assumption of convexity has the following implications regarding the level sets of F at $\pm\infty$. If F attains $-\infty$ at $u \in C$, then along every line segment ℓ in C having u as an endpoint, either F is constant at $-\infty$, or else there exists $v \in \ell$ such that F is constant at $-\infty$ up to v, F takes any value at v, and then F is constant at ∞ on the remainder of ℓ.

The search for solutions of optimization problems of arbitrary convex functionals seems a rather hopeless endeavor. If we are to have any hope of making the optimization problem tractable, we should impose some additional reasonable hypothesis on F.

Definition A.5. We say the convex functional F is **closed** if it satisfies the following properties:

1. F is *proper*, i.e., for all $u \in C$, $F(u) \neq -\infty$, and there exists $v \in C$ such that $F(v) \neq \infty$.
2. F is *lower semicontinuous*, i.e., for all $u \in C$,
$$\liminf_{v \to u} F(v) \geq F(u).$$

Remark A.6.

1. The assumption that F is proper ensures an optimal solution u^* of F, if it exists, is nontrivial, i.e., $u^* \notin \{\pm\infty\}$.
2. The assumption of lower semicontinuity is nontrivial because we allow F to take the value ∞. It is generally the minimal regularity assumption necessary to allow us to leverage the local behaviour of F in search for optimal solutions.

[4] In fact, the functional F we consider in the paper (Equation 4.13) is bounded, but we present the theory of convex analysis in this greater generality for completeness.

Proposition A.7 (Uniqueness of solutions)**.** *Suppose F is closed and strictly convex. Then there exists at most one solution to the optimization problem for F.*

Proof. In fact, we only need the assumption that F is proper. Suppose for a contradiction there exists $u, u' \in \operatorname{argmin}_{u \in C} F(u)$ distinct solutions. Let $F(u) = m = F(u')$. By strict convexity, for any $\lambda \in (0, 1)$,

$$F(\lambda u + (1-\lambda)u') < \lambda F(u) + (1-\lambda)F(u') = m$$

hence F does not attain a minimum at u or at u', a contradiction. □

Just as the derivative characterizes the existence of optima of convex functions on Euclidean space, there is an analogous result in the infinite-dimensional setting, given an appropriate notion of the derivative of F.

Definition A.8. Given $u, v \in X$, the **directional derivative of F at u in the direction** v, denoted $F'(u; v)$, is the limit defined by

$$F'(u; v) \coloneqq \lim_{h \to 0^+} \frac{F(u + hv) - F(u)}{h}.$$

Definition A.9. We say F is **Gâteaux differentiable at** $u \in X$ if for every $v \in X$, the directional derivative $F'(u; v)$ exists, and furthermore there exists a bounded linear functional $F'(u) \in X^*$ such that

$$\langle v, F'(u) \rangle = F'(u; v)$$

for all $v \in X$. We call $F'(u)$ the **Gâteux derivative of F at** u.

Remark A.10 (Relation with other notions of differentiability)**.** The Gâteux derivative generalizes the Fréchet derivative–the latter, if it exists, can easily be shown to satisfy the properties of the Gâteux derivative. The converse is *not* true in general.[5]

Proposition A.11 (First-order Conditions)**.** *Suppose F satisfies the following properties:*

1. *The functional F is closed.*
2. *The functional F is Gâteaux-differentiable in C with continuous derivative F'.*

Then, given $u^* \in C$, the following are equivalent:

1. *The element u^* is a solution of the optimization problem for F.*
2. *We have*

$$\langle v - u^*, F'(u^*) \rangle \geq 0 \tag{A.12}$$

for all $v \in C$.

Proof. First suppose $u^* \in C$ is a solution of the optimization problem for F. Given $v \in C$ and $0 < h < 1$, $h(v - u^*) \in C$ by convexity, and by minimality of $F(u^*)$,

$$\frac{F(u^* + h(v - u^*)) - F(u^*)}{h} \geq \frac{F(u^*) - F(u^*)}{h} = 0.$$

Taking $h \to 0^+$, we conclude

$$0 \leq F'(u^*; v - u^*) = \langle v - u^*, F'(u) \rangle.$$

Conversely, suppose Equation A.12 holds for all $v \in C$. Given $0 < h < 1$, by convexity of F, we have

$$\frac{F((1-h)u^* + hv)}{h} \leq \frac{1}{h} F(u^*) + F(v) - F(u^*).$$

Taking $h \to 0^+$, we get

$$\langle v - u, F'(u) \rangle = F(v) - F(u^*).$$

Since the left hand side is bounded below by 0 by assumption, and $v \in C$ is arbitrary, we conclude that $u^* \in \operatorname*{argmin}_{u \in C} F(u)$. □

[5] As in finite-dimensional calculus, the existence of the (total) derivative requires existence *and continuity* of the partial derivatives.

Appendix B. The Expectation-Maximization (EM) algorithm

B.1 Motivating and deriving EM

Our description and implementation of the EM algorithm closely follows Chapter 13 of Bishop [Bis06]. The major discrepancy (and where our implementation diverges from that of [AAJ21]) is that we work with log probabilities so as to mitigate underflow resulting from computing joint probabilities [6]

As described in Section 6.3, given a backtest $b \in [B]$, we have a time series $\boldsymbol{R} = R(b) \coloneqq \{r_n\}_{n=0}^{N-1}$ of N daily log returns corresponding to a sample of our observation process at the time points $\mathcal{T} = \{t_0 + n\Delta t\}_{n=0}^{N-1}$. For each $0 \leq n < N$, the n^{th} return r_n is (sampled from) a K-component Gaussian mixture model (GMM):

$$r_n \sim \sum_{k=1}^{K} \Theta_t^k N(\Delta t \gamma^{(k)}, \Delta t \Sigma),$$

where the mixing coefficients are determined by the states $\boldsymbol{\Theta} = \Theta(b) \coloneqq \{\Theta_n\}_{n=0}^{N-1}$ of the latent (π, Z)-Markov chain underlying the HMM.

The HMM property in continuous time implies the following discrete-time analogue for the process $(r_n, \Theta_n)_{n=0}^{N-1}$. At time $0 \leq n < N$, the log return r_n, conditioned on the latent state Θ_n, is independent of all other observed states and all latent states:

$$\mathbf{E}[r_n | \boldsymbol{R} \setminus \{r_n\}, \boldsymbol{\Theta}] = \mathbf{E}[r_n | \Theta_t].$$

In particular, via the chain rule for probability, this implies we can express the joint density of observed and hidden states as

$$\begin{aligned} p(r_0, \ldots, r_{N-1}, \Theta_0, \ldots, \Theta_{N-1}) &= p(\Theta_0) \left(\prod_{n=1}^{N-2} p(\Theta_{n+1} | \Theta_1, \ldots, \Theta_n) \right) \left(\prod_{n=0}^{N-1} p(r_n | r_1, \ldots, r_{n-1}, \boldsymbol{\Theta}) \right) \\ &= p(\Theta_0) \left(\prod_{n=0}^{N-2} p(\Theta_{n+1} | \Theta_n) \right) \prod_{n=0}^{N-1} p(r_n | \Theta_n) \end{aligned} \tag{B.1}$$

The first thing to notice regarding statistical inference of hidden Markov models is that the typical maximum likelihood estimate

$$\theta^{\text{MLE}} \coloneqq \operatorname*{argmax}_{\theta} \log \mathcal{L}^\theta,$$

where $\mathcal{L}^\theta \coloneqq p(\boldsymbol{R}|\Theta)$ denotes the likelihood of the observation sample $\boldsymbol{R}$ under the parameter set $\theta = (\gamma, \Sigma, \pi, Z)$, is not computationally tractable. This is because the latent states $\boldsymbol{\Theta}$ are unobserved, and so to compute the marginal density $p(\boldsymbol{R}|\theta)$, we must sum the joint density $p(\boldsymbol{R}, \boldsymbol{\Theta}|\theta)$ iterating over the entire path space $\boldsymbol{\Theta} \in E_K^N$–an $O(K^N)$ computation. To resolve this difficulty, the EM algorithm outlines an iterative procedure for obtaining a local maximum likelihood estimate $\theta^* = \theta^{(M)}$ of the parameter set θ, with each iteration consisting of the following two steps:

1. (*Expectation (E-)step*). Given the previous estimate $\theta^{(m)}$ of the parameter set, we evaluate $Q(\theta|\theta^{(m)})$, the expected (complete-data) log likelihood with respect to the posterior distribution $p(\boldsymbol{\Theta}|\boldsymbol{R}, \Theta^{(m)})$ of the latent state variables $\boldsymbol{\Theta}$:

$$\begin{aligned} Q(\theta|\theta^{(m)}) &\coloneqq \mathbf{E}[\log p(\boldsymbol{R}, \boldsymbol{\Theta}|\theta) | \boldsymbol{R}, \theta^{(m)}] \\ &= \sum_{\boldsymbol{\Theta} \in E_K^N} p(\boldsymbol{\Theta}|\boldsymbol{R}, \theta^{(m)}) \log p(\boldsymbol{R}, \boldsymbol{\Theta}|\theta). \end{aligned} \tag{B.2}$$

2. (*Maximization (M-)step*). We update the parameters $\theta^{(m+1)}$ as those which maximize expected log likelihood $Q(\theta|\theta^{(m)})$ defined in the E-step:

$$\theta^{(m+1)} = \operatorname*{argmax}_{\theta} Q(\theta|\theta^{(m)}).$$

[6] In particular, this circumvents the need to employ Levinson's scaling method to compute the joint probabilities in the forward step of the BW algorithm, as outlined in [Dev85].

The EM algorithm terminates either when some prespecified number of parameter-update iterations have occurred (in our case $M = 10$), or when the difference in log likelihood of subsequent iterations falls below some prespecified threshold (in our case $\mathrm{tol} = 10^{-4}$).

B.2 Implementing the EM algorithm

In order to carry out the E-step 1 of the EM algorithm, we need a method for computing $Q(\theta, \theta^{(m)})$. Following the notation in Bishop [Bis06], given $n \in [N]$, we let

$$\gamma(\Theta_n) \coloneqq p(\Theta_n | \boldsymbol{R}, \theta^{(m)}) \quad \text{and} \quad \xi(\Theta_{n-1}, \Theta_n) \coloneqq p(\Theta_{n-1}, \Theta_n | \boldsymbol{R}, \theta^{(n)})$$

denote posterior distributions of the latent state variable Θ. We write

$$\gamma(\Theta_{nk} | \boldsymbol{R}, \theta^{(m)}) \coloneqq p(\Theta_n = k | \boldsymbol{R}, \theta^{(m)}), \quad k \in [K],$$

hence we can identify $\gamma(\boldsymbol{\Theta}) \coloneqq (\gamma(\Theta_t))_{n=0}^{N-1}$ with an $N \times K$-stochastic matrix $(\gamma(\Theta_{nk}))_{n,k} \in \mathbf{R}^{N \times K}$. We use similar notation for $\xi(\boldsymbol{\Theta}) \in \mathbf{R}^{(N-2) \times K \times K}$.

Using characterization B.1 of the joint distribution of the observed and hidden states (and suppressing the dependence on the parameter set θ for notational clarity), we can efficiently express the expected log likelihood $Q(\theta, \theta^{(m)})$ as

$$\begin{aligned}
Q(\theta, \theta^{(m)}) &= \mathbf{E}[\log p(\boldsymbol{R}, \boldsymbol{\Theta} | \theta) | \boldsymbol{R}, \theta^{(m)}] && \text{(B.3)} \\
&= \mathbf{E}[\log p(\Theta_0) | \boldsymbol{R}, \theta^{(m)}] + \sum_{n=0}^{N-2} \mathbf{E}[\log p(\Theta_{n+1} | \Theta_n) | \boldsymbol{R}, \theta^{(m)}] + \sum_{i=1}^{n-2} \mathbf{E}[\log p(r_n | \Theta_n) | \boldsymbol{R}, \theta^{(m)}] && \text{(B.4)} \\
&= \sum_{k=1}^{K} \gamma(\Theta_{0k}) \log \pi^k + \sum_{n=0}^{N-2} \sum_{j,k=1}^{K} \xi(\Theta_{nj}, \Theta_{(n+1)k}) \log Z^{jk} + \sum_{i=1}^{n-2} \gamma(\Theta_{nk}) \log p(r_n | \Theta_{nk}). && \text{(B.5)}
\end{aligned}$$

The E-step of the EM algorithm reduces to computing the posteriors γ and ξ, which is done via the *forward-backward algorithm*. Subsequently, the M-step maximizing $Q(\theta|\theta^{(m)})$ can be readily performed via the method of Lagrange multipliers applied to Equation B.5. The forward-backward algorithm and the parameter update (M-step) are outlined in detail via the Algorithms 1 and 2.

The EM algorithm requires a parameter initialization $\theta^{(0)} = (\gamma(0), \Sigma(0), \pi(0), Z(0))$. We initialize the starting distribution $\pi(0)$ and the rows of the transition matrix $Z(0)$ of the latent Markov state which generates $\boldsymbol{\Theta}$ by sampling a Dirichlet distribution of order K with parameter $\alpha = 1/K\mathbf{1} \in \mathbf{R}^K_{>0}$. We initialize the mean vectors $\gamma(0)$ under the various K latent states as the cluster centers produced from applying the K-means clustering algorithm to the observation sample $\boldsymbol{R}$. Finally, the covariance matrix $\Sigma\,(0)$ is initialized as the sample covariance of the observation sample.

In order to minimize the dependence of the local maximum likelihood obtained the EM algorithm, we execute the EM algorithm 10 times for each backtest. In each run, we resample the initial parameters, and we keep the parameter set θ^* of the best run, as measured by the log likelihood $\log \mathcal{L}^\theta$.

Notation B.6. Given $k \in [K]$, we write

$$\mathrm{row}_k(Z) \coloneqq Z^\top e_k \quad \text{and} \quad \mathrm{col}_k(Z) \coloneqq Z e_k$$

for the k^{th} row and k^{th} column of the transition matrix Z.

Algorithm 1 Log forward-backward algorithm for the E-step of the EM algorithm

1. Compute the log Gaussian emissions $\log \varphi_n^k = \log p(r_n | \Theta_n = k)$ for $n = 0, \ldots, N-1$ and $k = 1, \ldots, K$.
2. Compute $\log \alpha$ and $\log P$ via the forward algorithm:
   1: Initialize $\log \alpha_0 = \log \pi + \log \varphi_0$.
   2: Set $n = 1$.
   3: For $k = 1, \ldots, K$, set
   $$\log \alpha_n^k = \log \varphi_n^k + \mathrm{LogSumExp}\left(\log \alpha_{n-1} + \log \mathrm{col}_k(Z)\right).$$
   4: Set $n \leftarrow n + 1$. If $n \leq N - 1$, go to step (2), else go to step (5).
   5: Set
   $$\log P = \sum_{k=1}^{K} \alpha_{N-1}^k.$$
2. Compute $\log \beta$ via the backward algorithm:
   1: Initialize $\log \beta^{N-1} = 0 \in \mathbf{R}^K$.
   2: Set $n = N - 2$.
   3: For $k = 1, \ldots, K$, set
   $$\log \beta_n^k = \mathrm{LogSumExp}(\log \mathrm{row}_k(Z) + \log \beta_{n+1} + \log \varphi_n).$$
   4: Set $n \leftarrow n - 1$. If $n \geq 0$, go to step (2).
3. Compute the log posteriors $\log \gamma$ and $\log \xi$:
   1: For $n = 0, \ldots, N-1$, set
   $$\log \gamma(\Theta_n) = \log \alpha_n + \log \beta_n - \log P.$$
   2: For $n = 1, \ldots, N-1$ and $k = 1, \ldots, K$, set
   $$\log \xi(\Theta_{(n-1)k}, \Theta_n) = \log \alpha_{n-1}^k + \log \varphi_n + \mathrm{LogSumExp}(\mathrm{row}_k\, Z + \log \beta_n) - \log P.$$

Algorithm 2 Parameter update (M-step) of the EM algorithm

1: Set
$$\pi = \gamma(\Theta_1) \Big/ \sum_{k=1}^{K} \gamma(\Theta_{1k}).$$
2: For $k, l = 1, \ldots, K$, set
$$A_{kl} = \sum_{n=1}^{N-1} \xi(\Theta_{(n-1)k}, \Theta_{nl}) \Big/ \sum_{k=1}^{K} \sum_{n=1}^{N-1} \xi(\Theta_{(n-1)k}, \Theta_{nl}).$$
3: For $k = 1, \ldots, K$, set
$$\gamma^{(k)} = \sum_{n=0}^{N-1} \gamma(\Theta_{nk}) r_n \Big/ \sum_{n=0}^{N-1} \gamma(\Theta_{nk}).$$
4: Set
$$\Sigma = \sum_{k=1}^{K} \sum_{n=0}^{N-1} \gamma(\Theta_{nk})(r_n - \gamma^{(k)})(r_n - \gamma^{(k)})^\top \Big/ \sum_{n=0}^{N-1} \sum_{k=1}^{K} \gamma(\Theta_{nk}).$$

References

[AAJ18] A. Al-Aradi and S. Jaimungal. Outperformance and tracking: dynamic asset allocation for active and passive portfolio management. *Appl. Math. Finance*, 25(3):268–294, 2018.

[AAJ21] A. Al-Aradi and S. Jaimungal. Active and passive portfolio management with latent factors. *Quant. Finance*, 21(9):1437–1459, 2021.

[APW14] A. Ang, D. Papanikolaou, and M. M. Westerfield. Portfolio choice with illiquid assets. *Management Science*, 60(11):2737–2761, 2014.

[Bis06] Christopher M. Bishop. *Pattern recognition and machine learning*. Information Science and Statistics. Springer, New York, 2006.

[Bro97] S. Browne. Survival and growth with a liability: Optimal portfolio strategies in continuous time. *Mathematics of Operations Research*, 22(2):468–493, 1997.

[Bro99a] Sid Browne. Beating a moving target: Optimal portfolio strategies for outperforming a stochastic benchmark. *Finance and Stochastics*, 3:275–294, 1999.

[Bro99b] Sid Browne. Reaching goals by a deadline: digital options and continuous-time active portfolio management. *Advances in Applied Probability*, 31(2):551–577, 1999.

[BSEJM08] C. Blanchet-Scalliet, N. El Karoui, M. Jeanblanc, and L. Martellini. Optimal investment decisions when time-horizon is uncertain. *Journal of Mathematical Economics*, 44(11):1100–1113, 2008.

[CM08] A. Celisse and A. Marrel. Selecting hidden markov model state number with cross-validated likelihood. *Computational Statistics*, 23(4):543–564, 2008.

[CSW24] Steven Campbell, Qien Song, and Ting-Kam Leonard Wong. Macroscopic properties of equity markets: stylized facts and portfolio performance, 2024.

[Dev85] Pierre A Devijver. Baum's forward-backward algorithm revisited. *Pattern Recognition Letters*, 3(6):369–373, 1985.

[DGU07] Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy? *The Review of Financial Studies*, 22(5):1915–1953, December 2007. _eprint: https://academic.oup.com/rfs/article-pdf/22/5/1915/24429471/hhm075.pdf.

[DN90] M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990.

[ET99] Ivar Ekeland and Roger Témam. *Convex analysis and variational problems*, volume 28 of *Classics in Applied Mathematics*. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, english edition, 1999. Translated from the French.

[Fer02] E. Robert Fernholz. *Stochastic portfolio theory*, volume 48 of *Applications of Mathematics (New York)*. Springer-Verlag, New York, 2002. Stochastic Modelling and Applied Probability.

[Mar52] Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[Mer69] R. C. Merton. Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case. *The Review of Economics and Statistics*, 51(3):247–257, August 1969.

[Mer71] R. C. Merton. Optimum consumption and portfolio rules in a continuous-time model. *J. Econom. Theory*, 3(4):373–413, 1971.

[MLZ22] Elmira Mynbayeva, John D. Lamb, and Yuan Zhao. Why estimation alone causes markowitz portfolio selection to fail and what we might do about it. *European Journal of Operational Research*, 301(2):694–707, 2022.

[Ode15] Gianluca Oderda. Stochastic portfolio theory optimization and the origin of rule-based investing. *Quantitative Finance*, 15(8):1259–1266, 2015.

[RW00] L. C. G. Rogers and D. Williams. *Diffusions, Markov processes, and martingales. Vol. 2*. Cambridge Mathematical Library. Cambridge University Press, Cambridge, 2000. Itô calculus, Reprint of the second (1994) edition.

[SS11] Elias M. Stein and Rami Shakarchi. *Functional analysis*, volume 4 of *Princeton Lectures in Analysis*. Princeton University Press, Princeton, NJ, 2011. Introduction to further topics in analysis.